WHAT DOES A JEPA WORLD MODEL PRESERVE?

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ABSTRACT

Joint-embedding world models learn both which variables to retain and how to predict them. These choices can fail in different ways: predictable nuisance can displace useful state, and a correct mean future can omit information needed for a decision. We study these requirements in minimal redcube worlds, moving from deterministic goal-directed motion to noisy observations and stochastic dynamics. Two exact examples expose a limitation even under ideal prediction. A planner can rank actions incorrectly when it evaluates squared goal distance at the mean rather than averaging outcome costs; recursively applying an exact nonlinear conditional mean can also miss the true multi-step mean. A known Gaussian nuisance construction gives the corresponding representation-level failure. In a matched three-seed study, the ranking of background persistence reverses with dynamics: per-frame noise gives lower position error than persistent noise without wind (0.115 versus 0.764 pixels), but much higher error with wind (21.33 versus 0.668). The same latent model supports nearby goal-directed movement, while the unchanged short-horizon controller degrades sharply on distant visual goals. Exact finite-branch calculations demonstrate small mean-only action-ranking changes in two frozen encoders, without a demonstrated physical-cost benefit. The central distinction is between retaining the state a task needs and retaining the predictive information its decisions need.

1 INTRODUCTION

In the tasks studied here, background texture changes neither the requested goal nor the object's dynamics or reward. A useful abstraction can ignore that texture while preserving where the object is and what an action can do to it. Joint-embedding predictive architectures (JEPAs) learn this abstraction by predicting future representations rather than reconstructing every future pixel (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024; Maes et al., 2026). The promise is attractive for physical intelligence: learn a compact description of the world that can be reused across goals. The difficulty is that predictive learning must decide both *what to represent* and *what kind of prediction a decision needs*.

These are separate requirements. First, an easy-to-predict feature need not be useful for control. A background fixed within an episode is perfectly predictable even if the controlled object is ignored (Sobal et al., 2022). Second, retaining the right state does not make its mean future a sufficient description of uncertain outcomes. An action that lands equally often on either side of a target has its mean exactly on the target. A planner that scores that mean can prefer it to an action that reliably lands close by. The mean prediction can be correct and the decision wrong.

We study both requirements in a minimal visual world: one red cube, four movement actions, a background, and optional stochastic wind or a moving wall. The simplicity is deliberate. Cube position, the requested goal, and the executed path are directly visible, so we can distinguish three cases without relying on a large game's aggregate score:

1. **Readable state and goal-directed movement.** Deterministic movement with changing visual noise is the initial control condition. Requested visual goals and actual trajectories test planning beyond merely staying alive.
2. **Failing to expose the controlled state.** Varying background persistence and wind tests whether the same predictive-learning recipe retains an accessible description of the cube.

3. **Retaining state while predicting an uncertain future.** Stochastic movement and a late-revealed opening test the decisions made from a learned imagination model. Exact finite-branch calculations separately isolate the effect of replacing a distribution by its mean.

Two elementary calculations make the third case precise. The first separates the cost of a mean from the expected cost of an outcome. The second shows that recursively applying an exact one-step conditional mean need not predict a later mean. Both hold under ideal prediction, without an optimization failure. A Gaussian nuisance construction supplies the corresponding limit on the representation-learning objective. These mechanisms motivate the experiments; the experiments determine which failure is actually observed in each learned model.

The distinctive empirical result is a reversal in the effect of nuisance persistence. Under predictable motion, changing gray noise yields more accurate position read-out than persistent noise; with fresh wind innovations, changing noise fails while persistent noise preserves subpixel read-out. The architecture, latent dimension, training data size, updates, and three-model replication are matched across this comparison. We use actual goal-directed paths and a shared-policy JEPA/flow comparison to determine what this representation boundary means for behavior. The point is not to rediscover that prediction and control differ, but to identify when the same predictive-learning recipe changes which physical state it exposes.

2 Two ways prediction can miss what a decision needs

2.1 An exact mean can rank actions incorrectly

For a fixed encoder, squared prediction loss is minimized by the conditional mean $m(c, a) = \mathbb{E}[Z' \mid c, a]$, where c is the available history. A goal planner instead needs to compare outcome costs. For a fixed latent goal z_g,

$$\mathbb{E}\big[\|Z' - z_g\|^2 \mid c, a\big] = \|m(c, a) - z_g\|^2 + \mathrm{trCov}(Z' \mid c, a). \tag{1}$$

The first term scores the mean; the second records the spread around it. An action-dependent spread can change which action minimizes expected cost.

Consider the exact scalar example in Figure 1. The goal is zero. Action A produces $X' = -1$ or $+1$, each with probability $1/2$; action B produces $X' = 0.1$ deterministically. The two means are 0 and 0.1, so their squared distances to the goal are 0 and 0.01: a mean-goal planner chooses A. The correct expected costs are 1 and 0.01: B is better. No improved estimate of the same mean repairs this ranking. The omitted variance is exactly the missing quantity.

2.2 An exact one-step mean need not compose into a later mean

Let $X_{t+1} = X_t^2 + \xi_t$, with independent fair noise $\xi_t \in \{-1, +1\}$ and $X_0 = 0$. The exact conditional mean is $m(x) = x^2$, so repeated mean prediction remains zero. In the real process, X_1 is -1 or 1 and X_2 is 0 or 2, each fairly. Consequently, $\mathbb{E}X_2 = 1 \neq m(m(0)) = 0$. More generally,

$$\mathbb{E}[X_{t+1}] = (\mathbb{E}X_t)^2 + \mathrm{Var}(X_t), \tag{2}$$

while recursive mean substitution retains only the first term. This discrepancy occurs with an exact one-step predictor and finite-support noise.

These examples concern mean-state rollout and goal evaluation. A model that predicts expected costs or suitable value functions directly can preserve the needed expectations without generating samples (Wan et al., 2019; Grimm et al., 2021). A distributional world model offers another route: it can retain distinct outcomes and average their costs. For the single quadratic goal cost in Equation 1, the mean plus one scalar variance trace per action already suffices. Whether a fitted model achieves that benefit is an empirical question.

2.3 Non-collapse does not determine which feature is retained

With finite second moments, a fixed encoder, and an unrestricted predictor, the minimum squared prediction risk is the expected conditional latent variance:

$$\inf_g \mathbb{E}\|g(C, A) - Z'\|^2 = \mathbb{E}\,\mathrm{trCov}(Z' \mid C, A).$$

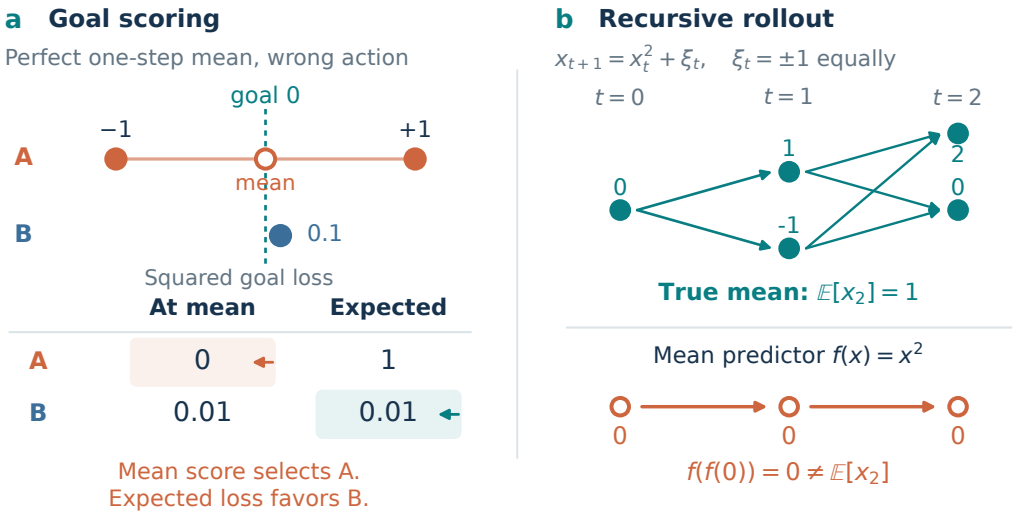


Figure 1: **Correct one-step means can still give wrong decisions and wrong rollouts.** Left: the exact mean of action A lies at the goal, although every outcome is farther from it than action B's deterministic outcome. Right: for $X_{t+1} = X_t^2 + \xi_t$, with independent fair $\xi_t \in \{-1, +1\}$ and $X_0 = 0$, repeated conditional-mean prediction stays at zero while the true two-step mean is 1. These are analytical examples, not outputs of trained image models.

Joint representation learning changes both the target and the conditioning features in this expression. It can make prediction easy by retaining a feature that barely changes. Write an observation as $X_t = R(P_t, B)$, where P_t is controlled state and B is an irrelevant nuisance fixed for the episode. If the nuisance is recoverable and can be mapped to $\phi(B) \sim \mathcal{N}(0, I)$, an encoder that returns only $\phi(B)$ is fully non-collapsed. Copying its last latent predicts the next one with zero error. A Gaussian marginal regularizer is also zero at the population level. Thus a global optimum can ignore the controlled state entirely.

This is the predictable-nuisance mechanism of Sobal et al. (2022), specialized here to the Gaussian-regularized objective. The construction assumes a sufficiently rich nuisance and a realizable encoder; it is not an assertion that a finite trained network reaches that optimum. It explains why non-collapse alone cannot choose the feature that matters. The empirical test is whether the learned representation continues to expose cube state when nuisance and dynamics change. Full assumptions and proofs are in Appendix B.

3 Minimal Worlds and Model Comparison

One object, controlled changes. A red 6×6 cube moves in a 63×63 RGB frame. Four native actions move it three pixels left, up, right, or down. Contact with the border terminates an episode with reward -1; other rewards are zero, and the cap is 128 steps. Background noise is either redrawn each frame or held fixed for the episode. Gray noise shares its sample across RGB channels. Optional wind is a hidden two-dimensional random walk: each coordinate increments by $-1, 0, +1$ with probabilities $0.1, 0.8, 0.1$, clipped to $[-2, 2]$, and is added to the action displacement. This produces fresh uncertainty even when previous motion is known.

The gate variant adds an approaching wall with a top or bottom opening sampled at spawn and revealed only after a fixed delay. Retreating gives time to observe the opening and respond. The common solid part of the wall is present in both possible futures, so its disappearance from a decoded forecast is an observed model/interface error, not the exact pixel average of the two gaps.

Joint-embedding prediction. The grid encoder produces $z_t = f_\theta(x_t) \in \mathbb{R}^{512}$, arranged as $4 \times 4 \times 32$ features. An action-conditioned, factored spatial–temporal predictor uses an eight-frame history. Training jointly updates encoder and predictor with coordinate-mean squared prediction loss and

SIGReg:

$$\mathcal{L} = \frac{1}{D}\widehat{\mathbb{E}}\|g_\theta(z_{\le t}, a_{\le t}) - z_{t+1}\|^2 + \lambda\,\mathrm{SIGReg}(\{z_t\}), \qquad D = 512. \tag{3}$$

There is no reward or reconstruction target in this representation-learning objective. The gray conditions share 500,000 training transitions and 8,500 updates, with three independently fitted JEPA seeds in the matched nuisance study. Decoders and state read-outs are fitted afterward on frozen encodings; their supervision and compute are additional to the world-model fit.

Distributional reference and behavior. The flow recipe uses a reconstruction-trained tokenizer of the same latent shape and a causal flow-matching predictor (Lipman et al., 2023; Hafner et al., 2025). It samples future latent states rather than replacing them by one point. Flow matching itself may use squared vector-field error; its sampled output distribution is the relevant distinction from a next-state mean. The model recipes differ in representation supervision, context, and task-head fitting. Their behavioral comparison therefore uses a shared pixel CNN and GRU, matched policy-training budgets, native actions/rewards, paired evaluation keys, and final checkpoints. In real evaluation, each policy observes posterior reconstructions from its own frozen model. Exact-simulator controls with the same decoded interfaces qualify what each policy can learn before imagination is assessed. Complete settings, replication levels, and earlier controls are in the appendix.

4 THREE CASES IN THE REDCUBE WORLD

4.1 CASE 1: A LEARNED LATENT MODEL CAN GUIDE NEARBY GOALS

With deterministic dynamics and per-frame gray noise, the grid JEPA exposes cube position with 0.115 ± 0.012 pixels of L1 error across three fits. We test whether its predictor and latent-goal cost can use that state. Horizon-three MPC enumerates all 64 native action sequences, minimizes summed squared distance to an encoded goal image, executes one action, and replans from the real observation. It uses no state coordinates, position head, decoder output, or transformed latent.

The model guides nearby goals reliably (Figure 2). All three fits reach every one- and two-action-distance target, and 94.5–96.9% of three-action-distance targets, within the common 24-action budget. Random actions reach 14.8% of the latter; horizon-one latent MPC reaches 12.5–17.2%. This establishes elementary goal-directed control through the learned representation and predictor despite changing visual noise.

The operating range is limited. Success falls to 19.5–26.6% at distance six, 0.8–2.3% at distance nine, and zero at distance 12. The original mixed six-to-12-action test remains a weak result: 5, 5, and 9 hits out of 128, with minimum-distance means worse than random. The full distance sweep was declared after that result and changes the goals, not the controller. It retains all six distances, all three models, and the same 128 valid histories. Thus the positive result is local goal-directed control, not long-horizon planning or a global guarantee about latent goal geometry.

The actual paths are selected by fixed case index, rather than success. Appendix O retains the original goal distribution, failures and complete secondary outcomes; Appendix O.1 gives the follow-up counts and selection rules.

4.2 CASE 2: NUISANCE AND STOCHASTIC MOTION CAN HIDE THE CONTROLLED OBJECT

Adding wind under the same per-frame gray background raises real-encoding position error to 21.33 ± 0.06 pixels across the three grid models. Keeping the background fixed for an episode instead gives 0.668 ± 0.049 pixels under wind (Table 1). This reverses the deterministic comparison, where per-frame noise gives lower error than persistent noise (0.115 versus 0.764 pixels). The observed boundary is therefore a noise-persistence $\times$ stochastic-motion interaction: persistent noise preserves accessible state under wind in the same recipe that fails with fresh noise.

The failure already appears when reading real encoded observations, before recursive prediction is evaluated. Reconstructions of those real encodings provide a complementary visual test of the learned observation interface. A failed read-out does not prove that every possible decoder lacks the state; it establishes a practical failure to expose position under the fixed diagnostic. The per-

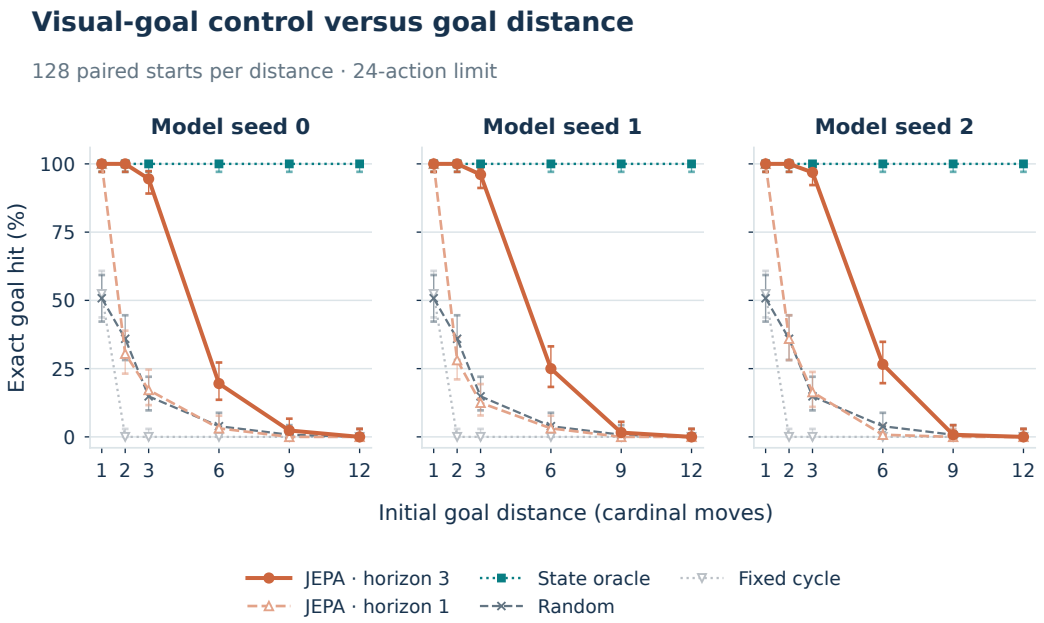


Figure 2: **The same frozen JEPA supports nearby goals and degrades beyond the short search range.** Exact first hit before death or 24 real actions, for each model and requested goal distance. Whiskers are per-cell Wilson 95% intervals over 128 cases; the repeated prefixes and model results are not pooled. This post-outcome diagnostic preserves the original failed six-to-12-action primary. The physical-state reference is privileged.

Table 1: **The same spatial JEPA exposes state in three conditions and fails the same read-out in the fourth.** Position L1 error in pixels, mean ± sample standard deviation across three model fits; architecture, 512-dimensional latent, training data size, and update count are fixed. "Real" uses encodings of held-out observations; predicted states use the same read-out eight steps ahead.

Gray background / dynamics	Real encoding	Eight-step prediction
Per-frame / deterministic	0.115 ± 0.012	4.84 ± 1.13
Per-frame / wind	21.330 ± 0.057	20.39 ± 0.92
Per-episode / deterministic	0.764 ± 0.068	1.83 ± 0.22
Per-episode / wind	0.668 ± 0.049	10.95 ± 0.53

frame/wind fits also retain high prediction loss, about 0.733 per coordinate. They are not examples of the zero-loss static-nuisance optimum from Section 2.3.

Internal draft: result pending. Case 2: insert the preselected true-frame / decoded-real / imagined-future panel and the completed shared-CNN behavioral results. Retain the persistent-noise comparison and all admitted policy seeds.

4.3 Case 3: Stochastic forecasts and imagined control

Persistent-gray wind is a candidate for separating representation from stochastic prediction: current position remains accessible, but future movement is uncertain. The delayed-reveal gate adds a decision that must wait for new information. In these settings, we compare policies trained inside JEPA and flow models through the same pixel-policy architecture, alongside real-simulator controls using the same posterior-decoded observations. This comparison tests the learned imagination packages; a claim that one model supports control requires the actual policy outcomes.

Internal draft: result pending. Case 3: insert the complete shared-CNN JEPA/flow comparison and decoded-real gate controls. State which interfaces qualify and which do not. Do not promote an unmatched historical flow success or attribute a gate gap specifically to conditional-mean averaging.

An exact nine-branch check on frozen encoders in black and persistent-gray wind worlds finds that discarding variance changes the optimal action for 2.65% and 1.08% of context–goal pairs under squared latent-goal cost, without a demonstrated improvement in physical goal cost when that variance is restored (Appendix P); this is a small direct illustration of the scoring operator, not a native control-success result.

5 Related work

Joint-embedding prediction avoids reconstructing all observable detail (Assran et al., 2023; Bardes et al., 2024). LeWorldModel jointly learns its encoder and action-conditioned predictor with squared loss and Gaussian regularization (Maes et al., 2026; Balestriero & LeCun, 2025); this is our direct learning setting. DINO-WM and V-JEPA 2 instead use frozen pretrained features for action-conditioned prediction, and PLDM jointly learns representations with additional regularizers (Zhou et al., 2024; Assran et al., 2025; Sobal et al., 2025). Sobal et al. already study action-conditioned moving objects with fixed or changing distractors, encoded and predicted state probes, repeated fits, and a generative reference (Sobal et al., 2022). Our distinct comparison adds fresh unobserved motion innovations to the matched spatial recipe: the effect of noise persistence reverses under wind. This is a boundary in a different recipe, not a contradiction of their findings. Recent contrastive work also examines slow-noise bias (Gulati & Nemenman, 2026).

Prediction and planning need not favor the same geometry. LeWorldModel reports accurate probes alongside weaker planning; Temporal-Distance JEPA and RC-aux introduce temporal or reachability constraints, while physical-state grounding adds task-state supervision (Maes et al., 2026; Bai & Xiong, 2026; Li et al., 2026; Yan et al., 2026; Liu et al., 2026). Terver et al. (2026) study architectural and planner choices and discuss multimodal conditional means. Our exact examples focus on the decision consequences of evaluating costs at a mean and recursively composing mean transitions.

FlowWM learns stochastic feature-space futures and evaluates mode coverage and future perception with a frozen visual encoder (Porcher et al., 2026). It is not action-conditioned and reports no control experiments. Our action-conditioned JEPA/flow comparison instead studies jointly learned state and real behavior. Generative imagination models have demonstrated useful control (Hafner et al., 2020; Alonso et al., 2024; Hafner et al., 2025), while also exhibiting failures of perception, action conditioning, and rollout (Hansen & Wang, 2026).

The broader prediction–control mismatch is longstanding (Lambert et al., 2020). Value-aware learning, expectation models, and value equivalence model the quantities needed by a particular decision class (Farahmand et al., 2017; Wan et al., 2019; Grimm et al., 2020; 2021). Temporal-difference JEPA predicts policy-conditioned successor features for value-based control (Bagatella et al., 2026). These approaches explain why our mean-state counterexamples are not impossibility results for all deterministic world models.

6 Discussion

The minimal worlds separate two demands on an abstract world model. Its representation must retain the variables on which a decision depends. Its prediction must preserve the expectations that the planner will evaluate. A non-collapsed representation can fail the first demand by concentrating on predictable nuisance. A correct conditional-mean model can fail the second when a planner applies nonlinear costs or composes nonlinear dynamics at that mean.

The redcube comparisons make these demands visible at different stages: requested-goal movement, position accessible in real observations, and behavior learned from imagined futures. The matched nuisance study already shows a reproducible change in accessible state under combined fresh noise and wind. Its high prediction loss separates the measured finite-budget failure from the idealized

nuisance optimum. Likewise, a practical JEPA/flow control gap would compare complete learned packages; the exact variance calculation is the narrower test that isolates point compression.

The empirical scope is small visual domains, specified models and training budgets, and the reported checkpoint and policy replications. The elementary examples establish limitations of mean-state planning, not a requirement that every useful world model generate samples. Expected-cost or value-equivalent models can preserve the needed decision quantities directly. General-purpose world models need a principle for choosing both the state to preserve and the predictive information that their intended decisions require.

Reproducibility statement

The appendices specify model fitting, post-hoc supervision, evaluation protocols, and replication levels. The tables retain final-checkpoint outcomes rather than selecting intermediate peaks. The accompanying evidence manifest identifies the source artifacts used for each result; independent policy seeds are distinguished from independent model fits. Claims based on a single checkpoint are labeled throughout.

Statement on the use of AI

Language models assisted with prose, code, plotting, proof checking, literature review, and adversarial manuscript review. The authors are responsible for all claims, experiments, and verification of the resulting text and artifacts.

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A WHAT THE OBJECTIVE DOES NOT GUARANTEE

The experiments concern fitted finite networks. The following population statements ask a narrower question: which guarantees follow from prediction error and a Gaussian latent marginal alone? They do not characterize optimization or identify the cause of the empirical failures.

A.1 NON-COLLAPSE DOES NOT SELECT THE CONTROLLED STATE

Write observations as $x_t = R(p_t, b)$, where p_t is controlled state and b is an episode-constant nuisance recoverable from the image. Assume that b is independent of states and actions in the training data. At deployment, assume it is independent of the initial state and transition randomness, and does not directly affect dynamics or reward. Let SIGReg_∞ be the nonnegative population characteristic-function discrepancy, which vanishes for $\mathcal{N}(0, I_D)$ (Appendix B).

Proposition 1 (Gaussian nuisance optimum; extending Sobal et al. (2022))**.** *If the encoder class contains a map $f(R(p, b)) = \varphi(b)$ with $\varphi(b) \sim \mathcal{N}(0, I_D)$ and the predictor class contains the last-latent copy map, their population prediction loss and SIGReg_∞ are both zero. This is a global optimum with covariance I_D and $I(z_t; p_{1:T}) = 0$ under the training distribution. A deployed controller observing only these latents, time, and its own actions has the value of a randomized state-blind policy.*

The argument is simple: nuisance changes across episodes, satisfying the marginal constraint, but is constant within each episode, making prediction exact. Its deployment value follows by giving a state-blind policy the same private nuisance draw and controller randomness. The Gaussianization assumption requires a rich nuisance and a realizable measurable map; a finite texture does not literally instantiate it. The finite-batch regularizer also differs from the population functional. A complementary innovation bound in Appendix B shows that an encoder preserving all state distances at a fixed scale must incur prediction error for unpredictable state innovations.

A.2 DECODABILITY DOES NOT SET A USEFUL PREDICTION SCALE

Even preserving all state information leaves its geometry unconstrained. For any $\varepsilon > 0$, we can place the two values of a relevant bit less than 2ε apart while keeping the *entire* latent distribution exactly standard Gaussian.

Proposition 2 (Exact information at arbitrarily small scale)**.** *For every $\varepsilon > 0$ and $D \geq 1$, there is a measurable observation process with episode-constant nuisance b and a fresh independent fair state bit p_t, and an encoder f_ε, such that $z_t \sim \mathcal{N}(0, I_D)$ and both p_t and b are exactly recoverable almost surely. For fixed b, $0 < \|f_\varepsilon(R(1, b)) - f_\varepsilon(R(0, b))\| < 2\varepsilon$. The optimal next-latent predictor is independent of the next bit and has coordinate-mean squared error in $(0, \varepsilon^2/D]$. With reward $\mathbf{1}\{a_t = p_t\}$, latent feedback attains value 1 per step, while every state-blind policy attains $1/2$.*

The construction partitions one Gaussian coordinate into narrow intervals and uses the bit to select the lower or upper half of each interval's conditional probability mass. The nuisance fixes the interval and the within-half quantile. This preserves the Gaussian marginal and exact decoding while making the two states arbitrarily close. The full proof is in Appendix B. It uses infinite-precision measurable coding; the next bit is intrinsically unpredictable. It therefore separates information from scale, without proving avoidable prediction error or explaining deletion of the predictable gate wall.

Why scale matters for a downstream read-out. For any fixed L-Lipschitz downstream score q (for example, an action-value read-out) and any coupled real and predicted next latents,

$$\left|\mathbb{E}q(Z') - \mathbb{E}q(\widehat{Z}')\right| \leq L\sqrt{\mathbb{E}\|Z' - \widehat{Z}'\|^2}. \tag{4}$$

This standard bound follows from Lipschitz continuity and Cauchy–Schwarz. Prediction error controls score error only relative to the read-out's sensitivity. Exact decodability alone supplies no bound on that sensitivity. We do not estimate L or claim a control guarantee for the fitted networks; the inequality makes explicit the missing bridge between a geometric loss and a decision.

B PROOFS

B.1 EMPIRICAL OBJECTIVE AND POPULATION IDEALIZATION

For a training batch of B windows and H predicted time positions, the implemented loss is

$$\mathcal{L}_{\text{pred}} = \frac{1}{BHD} \sum_{b,h} \|\widehat{z}_{b,h+1} - z_{b,h+1}\|^2. \tag{5}$$

The notation in Equation 3 averages over time and uses this coordinate-mean convention. One-step targets are teacher-forced at every temporal position, with a shorter history near the start of a window. For encoded windows of T frames, P random unit projection directions u_p, K knots $\tau_k \in [0, 3]$, and $w_k = e^{-\tau_k^2/2}$, the regularizer is

$$R = \frac{B}{TP} \sum_{t,p,k} q_k w_k \left[\left(\frac{1}{B} \sum_b \cos(\tau_k u_p^\top z_{b,t}) - w_k \right)^2 + \left(\frac{1}{B} \sum_b \sin(\tau_k u_p^\top z_{b,t}) \right)^2 \right], \tag{6}$$

where $\Delta = 3/(K-1)$, with $q_k = \Delta$ at the endpoints and 2Δ in the interior. The characteristic function is averaged across batch examples separately at each temporal position; the factor B is part of the implementation with $\lambda = 0.09$.

The population statements use the nonnegative discrepancy

$$\text{SIGReg}_\infty(\mathcal{L}(z)) = \int_{\mathbb{S}^{D-1}} \int_0^3 w(\tau) \left[\left(\mathbb{E}\cos(\tau\langle z, u\rangle) - w(\tau) \right)^2 + \left(\mathbb{E}\sin(\tau\langle z, u\rangle) \right)^2 \right] d\tau \, d\sigma(u), \tag{7}$$

where $w(\tau) = e^{-\tau^2/2}$ and σ is uniform on the sphere. Only the fact that this functional vanishes for a standard Gaussian is needed. It is not a theorem about exact minima of the finite-batch objective.

B.2 NUISANCE OPTIMUM AND THE COST OF INNOVATIONS

Proposition 1.

Proof. Let β recover the nuisance from the observation, set $f^\star = \varphi \circ \beta$, and let $g^\star$ copy the last latent. Then $z_t = \varphi(b) = z_{t+1}$, so every prediction residual vanishes. The Gaussian marginal gives covariance I_D and zero population regularizer. Both objective terms are nonnegative, so their sum is globally minimized at zero. Training-distribution independence gives $I(\varphi(b); p_{1:T}) = 0$.

For deployment, draw nuisance B and the controller's random seed U at reset, independently of the initial state and transition randomness. Run the original controller in the environment with nuisance B. Let a state-blind policy with private seed (B, U) feed constant latent history $\varphi(B)$ to a copy of that controller and reproduce its actions with U. Coupled to the same initial state and transition randomness, the policies take the same first action. If their actions agree through step t, their states, supplied latent histories, and seeds agree, so the next actions agree as well. Induction gives identical trajectories, returns, and terminations. This does not assume independence between B and states subsequently influenced by its induced actions. □

Corollary 1 (Innovation cost for uniformly state-resolving encoders)**.** *Let $p_{t+1} = p_t + \mu(a_t) + w_t$, $\mathcal{H}_t = \sigma(p_{\le t}, b, a_{\le t})$, and $\Sigma_t = \text{Cov}(w_t \mid \mathcal{H}_t)$. Assume finite second moments of states and encoded observations, $x_t = R(p_t, b)$, fixed nuisance b, and $\|f(R(p, b)) - f(R(p', b))\| \ge \kappa \|p - p'\|$ for all admissible p, p', b, with $\kappa > 0$. Every predictor of latent and action history satisfies*

$$\frac{1}{D}\mathbb{E}\|g - f(x_{t+1})\|^2 \ge \frac{\kappa^2}{D}\mathbb{E}\text{tr}\Sigma_t. \tag{8}$$

When $\mathbb{E}\text{tr}\Sigma_t > 0$, this exceeds the nuisance encoder's zero prediction loss under the assumptions of Proposition 1.

The assumption is stronger than injectivity or task sufficiency: every state distance must be preserved at a uniform scale. Innovation is measured after conditioning on state history and nuisance; persistent wind is not automatically fresh innovation.

Corollary 1.

Proof. Put $Z = f(x_{t+1})$ and $\mathcal{C}_t = \sigma(z_{\leq t}, a_{\leq t})$. Because observations and latents are measurable functions of state and nuisance, $\mathcal{C}_t \subseteq \mathcal{H}_t$. Any predictor using a shorter window is also $\mathcal{C}_t$-measurable. Conditional least squares and the tower property give

$$\begin{aligned}\mathbb{E}\|g - Z\|^2 &\geq \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &= \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{H}_t]\|^2 + \mathbb{E}\|\mathbb{E}[Z \mid \mathcal{H}_t] - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &\geq \mathbb{E}\mathrm{trCov}(Z \mid \mathcal{H}_t).\end{aligned}$$

The cross term vanishes because its first factor has conditional mean zero given $\mathcal{H}_t$. Take conditionally independent copies P, P' of p_{t+1} given $\mathcal{H}_t$. Such copies exist for Euclidean state spaces, using their regular conditional distributions. The nuisance b and current state p_t are fixed under this conditioning. The conditional independent-copy variance identity yields

$$\begin{aligned}\mathrm{trCov}(Z \mid \mathcal{H}_t) &= \tfrac{1}{2}\mathbb{E}[\|f(R(P,b)) - f(R(P',b))\|^2 \mid \mathcal{H}_t] \\ &\geq \tfrac{\kappa^2}{2}\mathbb{E}[\|P - P'\|^2 \mid \mathcal{H}_t] \\ &= \kappa^2 \mathrm{trCov}(p_{t+1} \mid \mathcal{H}_t) = \kappa^2 \mathrm{tr}\Sigma_t.\end{aligned}$$

The final equality follows because $p_t + \mu(a_t)$ is $\mathcal{H}_t$-measurable. Taking expectations and dividing by D proves Equation 8. Averaging these inequalities gives the corresponding bound for a multi-time prediction objective. Proposition 1 supplies the zero-loss comparator when its assumptions also hold. □

B.3 Conditional prediction and geometric scale

Lemma 1 (Conditional mean)**.** *For a fixed encoder and data law with $\mathbb{E}\|z'\|^2 < \infty$, the unique squared-error minimizer over measurable square-integrable predictors from history c and action a is $m(c,a) = \mathbb{E}[z' \mid c, a]$, up to null sets.*

Proof. Let $m = \mathbb{E}[z' \mid c, a]$. For every square-integrable measurable $g(c,a)$, expand $g - z' = (g - m) + (m - z')$. The cross term has expectation zero since $\mathbb{E}[m - z' \mid c, a] = 0$. Hence

$$\mathbb{E}\|g - z'\|^2 = \mathbb{E}\|g - m\|^2 + \mathbb{E}\mathrm{trCov}(z' \mid c, a).$$

The second term does not depend on g. The first is nonnegative and vanishes exactly when $g = m$ almost surely. □

The finite-sample regularizer. The finite-sample regularizer differs from Equation 7. For independent identically distributed samples with characteristic function ϕ and empirical estimate $\widehat{\phi}_n$,

$$\mathbb{E}|\widehat{\phi}_n(t) - w(t)|^2 = |\phi(t) - w(t)|^2 + \frac{1 - |\phi(t)|^2}{n}. \tag{9}$$

Indeed, $\mathbb{E}\widehat{\phi}_n = \phi$ and $\mathbb{E}|\widehat{\phi}_n - \phi|^2 = (1 - |\phi|^2)/n$, since each complex exponential has modulus one; expanding around ϕ gives the identity. At a Gaussian law, the residual equals $(1 - e^{-t^2})/n$. This is estimator bias, not an optimization floor. Changing the law changes both terms; dependent temporal samples introduce additional covariance terms. The implementation multiplies this statistic by batch size B, so its Gaussian finite-batch bias does not shrink as $1/B$ after that scaling. Finite projections and knots also do not characterize Gaussianity.

Proposition 2.

Proof. Partition $\mathbb{R}$ into $I_j = [2\varepsilon j, 2\varepsilon(j+1))$, $j \in \mathbb{Z}$. Let $q_j = \Pr(G \in I_j)$ for $G \sim \mathcal{N}(0,1)$. Every q_j is positive and their sum is one. Let F_j be the conditional Gaussian distribution function on I_j, with inverse quantile function Q_j on $(0,1)$. The conditional CDF is continuous and strictly increasing on the interval interior, so its inverse is measurable and takes values in that interior.

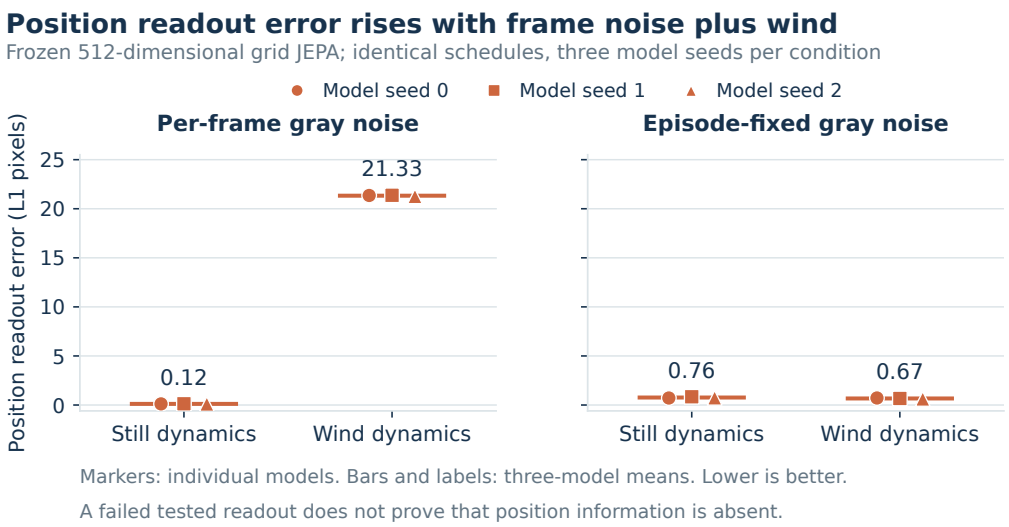


Figure 3: Matched gray-noise cube position read-out on real encodings. Each marker is one of three independently fitted grid JEPAs. Architecture, latent dimension, training data size, and update count are fixed; the displayed failure is specific to the fitted read-out and training recipe.

Draw J with probabilities q_j, $U \sim \mathrm{Uniform}(0,1)$, and $V \sim \mathcal{N}(0, I_{D-1})$, independently. Set $b = (J, U, V)$; for $D = 1$, omit V. Take $R(p, b) = (p, b)$ and define

$$f_\varepsilon(R(p,b)) = \big(Q_J((U+p)/2), V\big). \tag{10}$$

The mixture $(U + p_t)/2$ is uniform on $(0, 1)$, since p_t is an independent fair bit. Conditional on $J = j$, the first coordinate therefore has the Gaussian law restricted to I_j. Mixing with weights q_j recovers $\mathcal{N}(0, 1)$. Independence of V gives the joint law $\mathcal{N}(0, I_D)$ and zero population regularizer.

Given the first coordinate y, its interval determines J almost surely. Write $v = F_J(y)$. Then $p_t = \mathbf{1}\{v > 1/2\}$ and $U = 2v - p_t$ almost surely; the other coordinates recover V. Thus the latent recovers both state and nuisance. For fixed b, the two first coordinates are distinct and lie inside the same interval of width 2ε. All remaining coordinates agree, proving the distance bound.

Latent history reveals b, while the next bit remains independent and fair. The next latent consequently has two equiprobable values, say $z^{(0)}(b)$ and $z^{(1)}(b)$. Lemma 1 gives $m(b) = (z^{(0)}(b) + z^{(1)}(b))/2$. Its conditional squared error is $\|z^{(1)}(b) - z^{(0)}(b)\|^2/4$, strictly positive and bounded by ε^2. The midpoint is square-integrable by Jensen's inequality, since both state encodings have finite second moments. Taking expectations and dividing by D proves the coordinate-mean loss bound, including strict positivity since the conditional error is positive almost surely. The midpoint depends only on b, so it is independent of p_{t+1}. Finally, decoding the current bit permits $a_t = p_t$. A state-blind action is independent of the current fair bit, giving success probability $1/2$. □

C THE CUBE WORLDS

The position-readout study crosses black, per-frame random, and per-episode random backgrounds with deterministic or hidden-wind dynamics. Color-noise and black-background datasets contain 100,000 transitions; the gray-noise conditions contain 500,000. Only the four gray grid-JEPA conditions have three independent model initializations. Table 1 and Figure 3 report that replicated comparison. Tables 2 and 3 retain seed-0 outcomes for the broader conditions, using the same canonical benchmark files throughout. Differences across dataset sizes and model recipes are not isolated nuisance effects.

Environment. The scene contains a red 6×6 object in a 63×63 RGB image with a red boundary. Four cardinal actions move the object three pixels. Reset positions are uniformly sampled from

Table 2: Seed-0 cube position L1 error (pixels) for the grid and class-token JEPAs. "Real" evaluates held-out encoded frames; "Pred. 8" evaluates the same read-out on eight-step predicted latents. Gray conditions use more training transitions than black and color conditions.

Background	Wind	Grid JEPA		Class-token JEPA	
		Real	Pred. 8	Real	Pred. 8
Black	Off	0.097	3.933	0.087	45.268
	On	0.126	11.732	0.049	8.957
Color, frame	Off	0.172	5.570	24.133	19.428
	On	0.253	11.823	25.185	19.317
Color, episode	Off	3.273	6.594	20.804	21.567
	On	1.451	13.342	23.431	21.807
Gray, frame	Off	0.115	4.269	23.954	20.286
	On	21.354	19.696	25.288	19.567
Gray, episode	Off	0.715	1.779	21.316	21.284
	On	0.717	11.553	23.880	23.386

Table 3: Seed-0 flow cube position L1 error (pixels). Latent and pixel columns use different supervised read-outs. The persistent-color failure is visible before dynamics: real encoded position has large error. These are model-recipe references rather than objective-only comparisons.

Background	Wind	Real latent	Pred. 8 latent	Pred. 8 pixels
Black	Off	0.140	0.126	0.242
	On	0.153	11.632	11.622
Color, frame	Off	1.088	1.059	0.395
	On	1.344	12.311	11.985
Color, episode	Off	18.563	21.954	22.423
	On	7.010	18.960	16.937
Gray, frame	Off	3.114	2.667	0.382
	On	3.690	14.392	12.302
Gray, episode	Off	5.062	4.896	1.339
	On	3.066	15.150	13.583

integer coordinates in $[16, 42]^2$, and initial wind is zero. In wind conditions, each wind coordinate independently increments by -1, 0, or 1 with probabilities 0.1, 0.8, and 0.1, clipped to $[-2, 2]$; the updated wind is added to the action displacement. Background pixels are uniform integers from 0 to 255; gray noise shares the same sample across RGB channels. Contact with the boundary gives reward -1 and terminates; other rewards are zero, with a 128-step cap.

Read-out fitting. The nonlinear latent position probe is fitted on training data and evaluated on independent `eval.npz` frames. Its spatial softmax-and-offset output is produced by two hidden layers of width 128, trained for 10,000 updates with batch 2,048 and Adam at initial learning rate 10^{-3} with cosine decay. Grid inputs are standardized using training statistics. For class-token inputs, standardization and PCA whitening are selected using the final 10% of fitting rows, then the chosen variant is refitted on the full fitting split. Whitening floors singular values at 1% of the largest. Applying this read-out to predicted latents tests both prediction and read-out extrapolation. The pixel read-out used for decoded flow images is a separate supervised CNN trained with synthetic background replacement using ground-truth cube masks; it is a diagnostic, not a policy or a world-model training signal. Rollout windows are sampled with replacement and overlap, and exclude death or reset during the eight context and 16 forecast frames. The configured benchmark uses 2,048 windows for video metrics and the first 512 for position diagnostics; exact sampled indices were not retained in the archived result files. Surviving windows are a selected subset of wind trajectories, so these errors do not establish death prediction or closed-loop control.

Earlier controller experiments use the three black/color background conditions. Their final survival outcomes are retained below. The class-token JEPA's near-zero loss on per-episode texture and

poor reward-driven control are consistent with the known predictable-nuisance mechanism, without exactly realizing Proposition 1. The replicated gray-frame/wind failure instead has high prediction loss and should not be interpreted as the same mechanism. Still-world survival can be achieved by state-blind alternating actions, so it does not certify state feedback.

Table 4: Survival (percent) in the supplementary cube worlds. "LeWM" denotes the class-token JEPA; "grid" denotes the grid JEPA; "goal" and "heads" use latent-goal and learned-return search, respectively. PPO columns use the final saved policies; "px" and "z" identify pixel and latent observations. Still-world survival does not certify state feedback.

Background	Black		Per-frame noise		Per-episode noise	
Dynamics	Still	Wind	Still	Wind	Still	Wind
random	20	0	20	0	20	0
centre	100	99	100	99	100	99
LeWM goal	97	64	5	0	80	0
LeWM heads	100	60	9	0	2	0
LeWM PPO	100	99	0	0	0	0
grid goal	88	1	92	0	100	77
grid heads	100	5	100	0	85	8
grid PPO	100	94	100	0	2	0
flow PPO px	100	94	98	97	98	95
flow PPO z	100	96	99	1	51	1
PPO real	100	98	100	96	100	78

D HYPERPARAMETERS AND COMPUTE

The gate and gray-noise cube conditions supply 500,000 uniform-random transitions each. The earlier black and color-noise cube datasets contain 100,000 transitions. Craftax uses its separate 500,000-transition archive. Model training uses bfloat16 compute, float32 parameters, and one A100. A 500-step calibration estimates the training length fitting 575 seconds including compilation; the grid gate model trains for 8,500 steps (17,000 and 34,000 in the earlier budget extensions of Appendix J), the class-token cube models for 23,750–24,000. Model seeds 0, 1, 2 of the grid JEPA differ only in the initialisation and data-order seed.

JEPA settings. Both JEPA architectures use batch 128 and AdamW with learning rate and weight decay 10^{-3}, 10% warm-up followed by cosine decay to zero, and gradient clipping at 1.0. SIGReg uses weight 0.09, 17 knots on $[0, 3]$, and 1024 random projections. Neither architecture uses a stop-gradient, teacher, decoder, or reward during world-model fitting. Linear–BatchNorm–GELU–linear projection heads flatten batch, time, and spatial tokens for normalization. Training-mode statistics therefore couple examples and temporal positions; at inference, running statistics define a fixed pointwise projection and the attention predictor is causal. The theoretical causal-map formulation is an inference/population idealization.

	Class-token JEPA	Grid JEPA
encoder	ViT, 7×7 patches	tile transformer + conv
width / depth / heads	192 / 6 / 3	128 / 6 / 8
encoder MLP	768	$4\times$ width
read-out	class token	4×4 grid, 32 channels
latent dimension	192	512
projector	BN projector, width 2048	same, hidden 512
predictor	AdaLN-zero AR	spatial + causal temporal
predictor depth / heads / dim	6 / 16 / 64	6 / 8 / 32
predictor MLP / width	2048 / latent	$2\times$ width / 256
history	3 frames	8 frames
dropout	0.1	0
parameters	15.3M	2.6M + 8.3M

Flow settings. The tokenizer uses 9,000 steps, batch 256, and learning rate 9×10^{-4}; dynamics 16,000 steps, batch 64, the same learning rate, width 256, six layers, eight spatial and temporal attention heads, a 32-frame context, independent per-frame noise levels, a shortcut loss of weight 0.25 every fourth step after the first quarter of training, history dropout 0.25, two history frames resampled every eight steps after half of training, and a temporal loss weight of 0.25. The symlog reward head has 255 bins over $[-20, 20]$; reward and termination losses have weight 0.1 and enter the trunk at full strength for half of training, then at factor 0.1. Inference uses four function evaluations per frame. We corrected terminal-transition masking in the recipe's reward loss (Appendix H).

JEPA read-outs. Gate post-hoc heads use the frozen encoder and the first 800 time rows (80,000 frames) of the same archive used for world-model training. The following 200 time rows are held out from head fitting only; the independent world-model evaluation archive is `eval.npz`. The pixel decoder uses 8,000 steps, batch 256, learning rate 10^{-3}, width 256, six residual blocks; it supports the figures and the decoded-latent control, in which its frozen output, clipped to $[0, 255]$, replaces the pixels of the real-gate PPO; no true pixel enters that policy after the encoder. Reward, termination, and behaviour-value heads are two-layer perceptrons of width 512 on the latent and the one-hot action, 6,000 steps, batch 1024, learning rate 10^{-3}, Monte Carlo value targets at discount 0.99, value loss weight 0.1. The residual predictor models $z_{t+1} - m$ with $m = g(z_{t-7:t}, a_{t-7:t})$ the original grid prediction, conditioned on m, z_t, and the action: three residual blocks of width 1024, 8,000 steps, and batch 2048. The final 20,000 valid windows of the same world-model training archive are excluded from residual fitting and used for its quality check. They are not independent world-model evaluation data. For encoder seed 0, coordinate-mean MSE is 3.07×10^{-4} for the original predictor, 2.05×10^{-4} for one residual sample, and 1.32×10^{-4} for the mean of eight. This

quality check uses eight function evaluations per sample; the reported PPO runs instead use four. Averaging eight learned samples is a random Monte Carlo average, not an exact conditional mean.

Policy optimisation. PPO uses discount 0.99, GAE parameter 0.9, clipping 0.2, value coefficient 0.5, entropy coefficient 0.01, AdamW at 2.5×10^{-4} with weight decay 0.01, four epochs per batch, and gradient clipping at 1.0. Pixels pass through a three-block Impala torso (Espeholt et al., 2018), latents through a two-layer perceptron; a GRU receives that embedding and the previous action; the critic uses 64 categorical bins with histogram-Gaussian targets (Farebrother et al., 2024). Real-environment PPO uses 128 environments and 64 steps per iteration with four minibatch groups, truncated backpropagation over 16 steps, and 1.5M environment steps. In-model PPO uses 150 updates of 512 trajectories of 16 steps (1.2M imagined steps) with minibatch size 128; half of the starts are resets, half dataset states with their real eight-frame context. All reported policy rows use final checkpoints.

Compute. On a shared A100, the seed-0 grid-gate JEPA completes 8,500 updates in 573 seconds and has a logged total of 586.5 seconds. Probe and head fitting takes approximately three minutes. The residual sampler's optimization stage takes 74.3 seconds; model loading, latent extraction, fitting, and quality evaluation together take 275.2 seconds before the final file write. Other logged stages are PPO inside the JEPA 2.0–4.3 min per run; PPO inside the flow model 5–6 min per run at four function evaluations, 20–26 min with the mean of eight; PPO in the real environment 3.7–7.7 min on pixels with the convolutional torso and 51–70 s with the MLP torso, 76–78 s on the frozen latent, 4.7–4.8 min on the decoded latent with the convolutional torso and 1.8–2.9 min with the MLP torso; the latent diagnostic: 2 min.

E Evaluation protocol

Every gate-policy endpoint uses the final checkpoint and 256 real episodes from reset, with the 128-step cap. Survival is the fraction still alive after step 128; death on that step is not survival. Walls passed counts wall exits before or on the terminal transition, including an exit on the same step as death. Discovery is final survival strictly above 50%. Real-training and imagined-training runs advance their random keys under different iteration schedules; their final evaluations are not paired episode trials. Policy seeds are separate runs conditional on a fitted model; independently fitted encoders provide a different replication level. The flow model has only one checkpoint. Reported means weight policy runs equally; standard deviations describe variation across those runs, not uncertainty across model families.

Gate specification. The gate uses the same image and object sizes and cardinal action displacement. A white vertical wall of width three pixels spawns at $x = 60$ every 22 steps and moves left three pixels per step. Its 16-pixel gap is sampled at spawn, top or bottom with equal probability, and becomes visible when the wall reaches $x \leq 40$. The gap location is hidden state before that reveal. Border or wall contact produces reward -1 and termination; the time cap is 128 steps.

Latent diagnostic. For each encoder, the diagnostic uses 64 real contexts with the wall still solid at $x \in \{42, 45\}$. Reconstructed simulator states generate recorded futures and matched wall-free futures under the recorded actions. The recorded branch survives the diagnostic horizon by construction. At each step, wall amplitude is the squared Euclidean distance between encodings of the wall-present and wall-absent frames. The teacher-forced residual uses real encoded history and is a total squared error, summed over latent coordinates. The wall coordinate of a prediction is its projection onto the wall-present minus wall-absent latent difference, with endpoints 0 and 1. Decoded wall intensity averages the minimum colour channel on shared wall pixels, excluding either possible gap band. These measurements compare features within one encoder; raw distances across models are not an information metric.

F Latent-policy torso controls

The MLP on decoded latents passes 0.92 walls per episode with no discoveries in six runs (Table 5), using encoders 0 and 1. The decoded CNN control instead uses all three encoders, with three

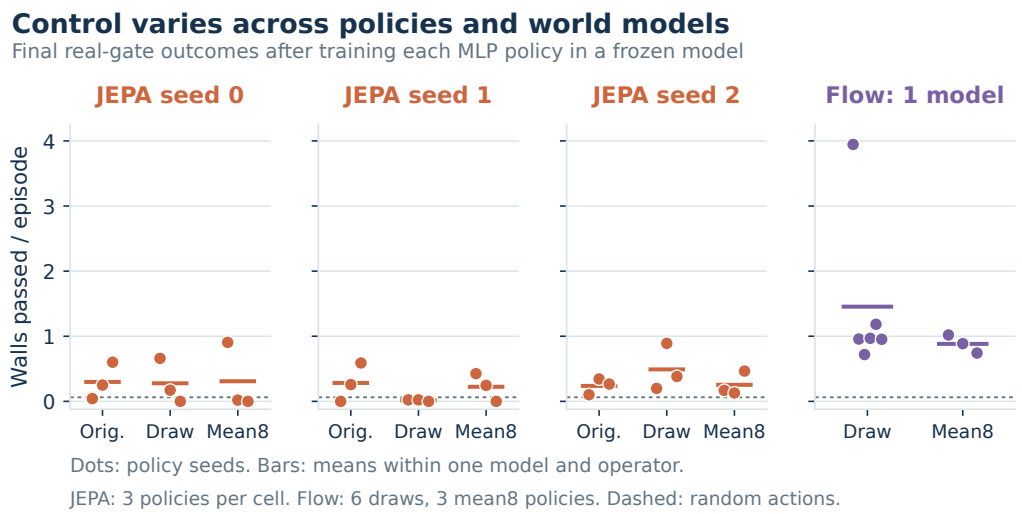


Figure 4: **Control outcomes by model fit and predictor variant.** Each point is a final policy evaluated on 256 real episodes. Original prediction, residual sampling, and residual means are separate conditions; model and policy replication are distinct. Horizontal bars show within-model means, and the dashed line marks random actions. The flow reference uses one checkpoint.

policies per encoder. Both of its discoveries use encoder 1. The decoder is frozen and reconstruction-supervised; success therefore establishes useful encoded information under this added read-out, not sufficiency of the unsupervised training objective. Failure of the true-pixel MLP recipe establishes no upper bound on MLP capacity.

Table 5: Real-gate observation and torso controls: final walls per episode and survival, mean ± standard deviation over policies; discovery means final survival above 50%. All policies use 1,499,136 real transitions.

observation of the real-gate PPO	runs	walls passed	survival (%)	discovered
pixels, convolutional torso	3	2.91±1.65	65±52	2/3
pixels, MLP torso	3	0.91±0.08	4±2	0/3
frozen JEPA latent, MLP torso	9	0.94±0.04	4±1	0/9
decoded JEPA latent, convolutional torso	9	1.57±1.26	22±37	2/9
decoded JEPA latent, MLP torso	6	0.92±0.08	3±2	0/6

G CONTROL OUTCOMES AND LATENT DIAGNOSTICS

Table 6 gives aggregate final outcomes; Table 7 lists every imagined JEPA policy. The pooled imagined JEPA mean weights all original, sampled-residual and mean-residual runs equally. The flow rows distinguish six sampled policies from three policies using a mean of eight, all sharing the same model fit. They do not estimate variation across independent flow-model fits. Tables 8 and 9 retain the encoder and geometry diagnostics supporting the gate observations.

Table 6: Final control outcomes in the real gate, mean ± standard deviation over policy runs. Imagined training uses 1,228,800 transitions per policy; real training uses 1,499,136. Model-fit counts and policy-run counts identify distinct replication levels.

Controller	Models	Policies	Walls	Survival (%)	Discovery
JEPA: original	3	9	0.27 ± 0.22	0 ± 1	0/9
JEPA: mean of 8	3	9	0.26 ± 0.30	1 ± 1	0/9
JEPA: one sample	3	9	0.26 ± 0.32	1 ± 1	0/9
Flow: mean of 8	1	3	0.88 ± 0.14	3 ± 1	0/3
Flow: one sample	1	6	1.45 ± 1.23	19 ± 39	1/6
Random	–	–	0.06	0	–
Alternating	–	–	0.02	0	–
Real pixels: CNN	–	3	2.91 ± 1.65	65 ± 52	2/3
Real pixels: MLP	–	3	0.91 ± 0.08	4 ± 2	0/3
Real JEPA: MLP	3	9	0.94 ± 0.04	4 ± 1	0/9
Real decoded: CNN	3	9	1.57 ± 1.26	22 ± 37	2/9

Table 7: Every imagined JEPA policy at its final checkpoint: survival (percent) and walls per episode by encoder seed, predictor variant and PPO seed. Each cell uses 256 real episodes.

		Survival (%)			Walls passed		
Predictor	Model	PPO 0	PPO 1	PPO 2	PPO 0	PPO 1	PPO 2
Original	0	0	0	1	0.04	0.25	0.60
	1	0	0	1	0.00	0.26	0.59
	2	0	1	0	0.10	0.34	0.27
Residual mean	0	3	0	0	0.91	0.02	0.00
	1	2	1	0	0.43	0.25	0.00
	2	0	0	1	0.17	0.13	0.46
Residual sample	0	2	1	0	0.66	0.17	0.00
	1	0	0	0	0.02	0.02	0.00
	2	0	3	1	0.20	0.89	0.38

Table 8: Grid-JEPA diagnostics by encoder seed. Wall amplitude and total teacher-forced residual are squared distances at future steps 1 and 8; decoded intensity uses steps 1 and 2. The wall coordinate has endpoints 0 (wall absent) and 1 (true encoding). Head AUC compares each death cause with alive transitions; PPO wall counts pool predictor variants.

JEPA encoder seed	0	1	2
Wall amplitude, step 1	0.035	0.028	0.088
Wall amplitude, step 8	0.124	0.082	0.193
Total residual, step 1	0.145	0.130	0.156
Total residual, step 8	0.249	0.228	0.279
Wall coordinate, step 1	0.39	0.45	0.66
Decoded wall, step 1	0.00	0.00	0.68
Decoded wall, step 2	0.00	0.00	0.02
Border-death AUC	0.96	0.96	0.88
Wall-death AUC	0.65	0.65	0.76
Imagined PPO, walls	0.29±0.34	0.17±0.22	0.33±0.24

Table 9: JEPA geometry and decoded wall intensity for model seed 0, averaged over 64 pre-reveal contexts. Distances use matched actions and the same encoder; total residuals include all latent coordinates. Wall intensity is measured on shared wall pixels.

Future step	1	3	5	8
Squared distance, summed over coordinates				
Wall-present versus wall-absent	0.035	0.041	0.071	0.124
Total teacher-forced residual	0.145	0.152	0.200	0.249
Residual along wall direction	0.015	0.021	0.048	0.104
Residual orthogonal to wall direction	0.130	0.132	0.152	0.145
Predicted wall coordinate	0.39	0.33	0.29	0.21
Decoded wall intensity				
Encoded true future	1.00	1.00	1.00	1.00
Original JEPA rollout	0.00	0.00	0.00	0.00

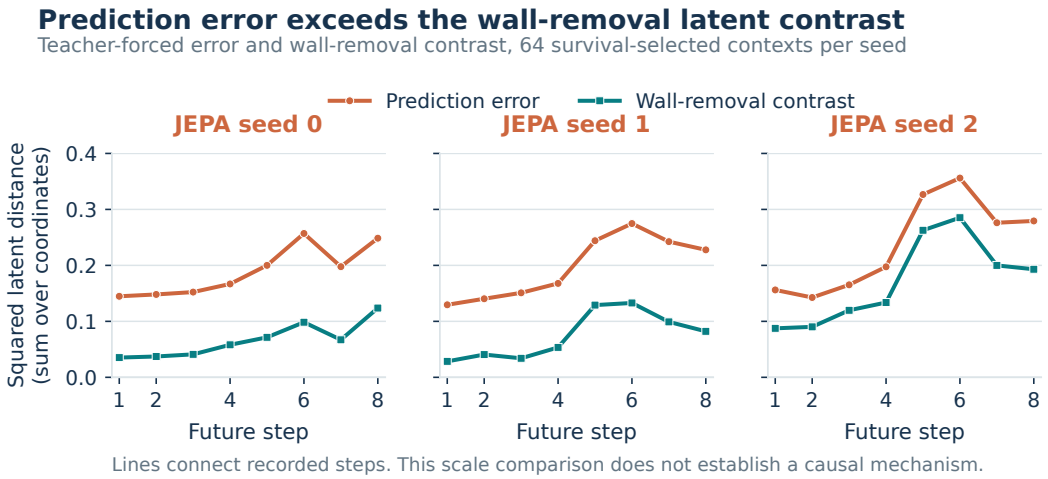


Figure 5: **Within-encoder scale diagnostics.** For each of the three gate encoders, wall-removal distance and teacher-forced total one-step residual are summed squared latent distances. Curves average 64 pre-reveal contexts whose recorded futures survive the diagnostic horizon. These are feature sensitivity and total prediction error, not commensurate measures of information content.

H TERMINAL REWARDS IN THE FLOW RECIPE

The original recipe masks transitions entering reset frames out of reward training. Here the only nonzero reward occurs on terminal transitions, so that mask removes the reward signal. The corrected head predicts the next reward from the current frame summary and action with terminal transitions retained in its loss. The reported flow-policy outcomes use this corrected convention. The single-checkpoint reference therefore denotes the complete tokenizer, dynamics, reward and termination package, rather than an isolated change in transition stochasticity.

I CRAFTAX PROTOCOLS AND SINGLE-CHECKPOINT RESULTS

I.1 FRESH LOCAL-GOAL EVALUATION

The new evaluation uses the same existing 512-dimensional checkpoint, model seed 0, with 128 starts for each evaluation seed 101, 202, and 303. Goal generation and real execution have separate pseudorandom streams; all controllers share initial states and the execution stream. The target is the endpoint after six random actions, following seven warm-up actions from reset. The objective is final player-position equality with survival throughout the six executed actions. Every evaluated trajectory survives, so the survival requirement does not alter the measured rates. The target position displacements are zero, one, two, or three grid cells in 137, 148, 84, and 15 cases, respectively.

The open-loop planner solves once at horizon six. Receding-horizon planning receives each real observation, executes the first selected action, and replans over the remaining horizons $6, 5, 4, 3, 2, 1$. Every CEM solve uses 512 candidate sequences, 32 elites, and six iterations. Thus receding planning uses six times as many candidate-sequence searches and 3.5 times as many predicted latent transitions as one-shot planning, excluding additional encoding. It is not a compute-matched feedback intervention. Both controllers use terminal latent MSE and the native 17 actions, without reward heads, decoder use, latent rescaling, or privileged coordinates in the planning cost.

Table 10: Craftax goal-success counts and Wilson 95% intervals, conditional on one fitted model. The reference replays the privileged sequence used to generate the goal under independent execution noise; it is a protocol check, not an oracle or a guaranteed upper bound.

Controller	Success / total	Rate (%)	Wilson interval (%)
No-op	$137/384$	35.68	$[31.05, 40.59]$
Random	$65/384$	16.93	$[13.51, 21.00]$
JEPA open loop	$358/384$	93.23	$[90.26, 95.34]$
JEPA receding horizon	$369/384$	96.09	$[93.66, 97.62]$
Reference-action replay	$382/384$	99.48	$[98.12, 99.86]$

Open-loop and receding-horizon success on displaced targets is $222/247$ and $232/247$; for displacements of at least two cells it is $80/99$ and $87/99$. Random actions reach $19/247$ and $1/99$ in these strata, while no-op reaches neither. Mean endpoint position L1 errors are 0.1068 open loop, 0.0781 receding horizon, 1.5599 random, and 0.9401 no-op, in grid-cell units. Receding-horizon success exceeds open loop by 2.86 percentage points; a paired episode bootstrap with 10,000 resamples gives $[0.78, 4.95]$ percentage points. All intervals quantify evaluation variation for this fixed checkpoint. The three evaluation seeds are not training replications.

Table 11: Craftax success by target displacement. The first column includes all 384 starts; the others retain 247 displaced targets and 99 targets at least two grid cells away.

Controller	All (384)	Displaced (247)	≥ 2 cells (99)
No-op	35.7%	0.0%	0.0%
Random	16.9%	7.7%	1.0%
JEPA open loop	93.2%	89.9%	80.8%
JEPA receding horizon	96.1%	93.9%	87.9%

I.2 Earlier exploratory goal and reward protocols

The earlier results below use different evaluation episodes and a shared goal-generation/execution noise protocol. They are not pooled with the fresh independent-noise evaluation above.

The grid checkpoint uses training seed 0, a tile encoder, $16 \times 32 = 512$ latent coordinates, a six-layer factored predictor of width 256, history length eight, 17 actions, batch 128, and 8,500 updates. The recorded parameter count is 11.12 million. Training takes 604.5 seconds from loading through the logged end-to-end timer (578.5 seconds in the update loop); thus it is an approximately ten-minute run, not a strict sub-600-second result. The frozen decoder and heads add 175.6 seconds. Results are from one model fit and evaluation seed 0.

Goal generation and planning. Reset states receive seven random warm-up actions. Six further random actions generate a reachable goal image. CEM optimizes a length-six action sequence in latent space, with population 512, 32 elites, and six iterations. The learned planner uses the goal image, without privileged position coordinates or the generating actions in its cost. It executes the entire selected sequence without observation feedback. Success is equality of final player positions. Goal generation, random actions, and the selected plan use the same future simulator random keys; replaying the generating actions therefore reaches all goals and serves only as a protocol check. This protocol samples nearby reachable goals, not arbitrary distant tasks.

The planned endpoint position L1 error is 0.0781, against 1.5313 for random actions. Pixel MSE on the 0–255 scale is 104.00 versus 1477.92. These quantities use the same 128 starts as the reported $121/128$ versus $24/128$ reach counts. They do not quantify variation across fitted models.

Reward control and head generalization. Reward and value heads use the first 80,000 transitions (320 time rows across 250 streams) from the world-model training archive. Held-out time rows are excluded from head fitting but not from encoder training. Value fitting uses the first 80 time rows owing to truncated-return handling. Reward and value fitting correlations are 0.922 and 0.995; held-out correlations are 0.353 and 0.266. Held-out reward RMSE is 0.222. Mean predicted held-out value is 2.356, versus mean empirical discounted return 1.348.

Reward evaluation uses 128 starts, 100 steps, no autoreset, and rewards accumulated while alive. The CEM controller executes one action and replans from each real observation. Random actions, myopic fitted-reward argmax, one predicted step plus value, and CEM obtain returns 1.4453, 0.7813, 0.9438, and 0.4734, respectively; their across-start standard deviations are 1.0897, 1.1109, 1.2014, and 1.1039. Fractions alive at the end are 0.6328, 0.7266, 0.6328, and 0.6719. Aggregate summaries do not preserve the per-start covariance needed for paired confidence intervals. The archive's mean reward multiplied by 100 is not an online behavior baseline and is excluded from the comparison.

Earlier Craftax experiments used a different 81×32 spatial latent and different horizons. They are not merged with these 512-dimensional results. There is no matched online flow-controller comparison for this checkpoint.

J Earlier unmatched gate experiments

Reference model. For the gate, a reconstruction-trained tokenizer and causal flow-matching predictor provide a feasibility reference (Lipman et al., 2023; Chen et al., 2024; Hafner et al., 2025). It uses the same latent shape, data, and nominal model-training budget, with a 32-frame context and jointly fitted reward and termination heads. Architecture, supervision, and history length differ from the JEPA. One dynamics checkpoint is evaluated, so its policy runs show variation conditional on that fit rather than variation across fitted flow models.

J.1 Useful wall information remains in the frozen representation

Decoded real encodings visibly retain the approaching wall (Figure 6). To test whether that information can support behavior, we train a convolutional policy in the real gate on the frozen decoder's output. Two of nine runs discover the survival strategy, both using encoder seed 1; the corresponding real-pixel CNN discovers it in two of three runs (Figure 7). This establishes useful information in at least one fitted encoder under the decoder-and-CNN read-out. The decoder adds reconstruction

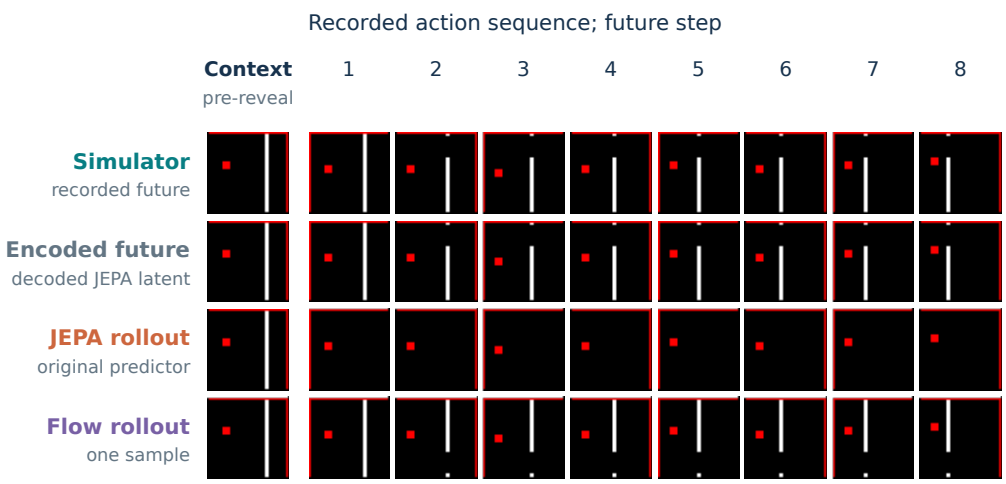


Figure 6: **A wall visible in real encodings disappears in predicted rollouts.** A pre-reveal context is followed by futures under recorded actions: the simulator, decoded real JEPA encodings, decoded original JEPA predictions, and one flow-model sample. The JEPA rows use encoder seed 0 and the same fitted decoder. The example is the first stored diagnostic context at wall position $x = 45$ among 64 contexts whose recorded futures survive the horizon. The flow sample is a single-checkpoint feasibility reference. A decoding failure on predicted latents can reflect both prediction error and off-distribution read-out; it is not alone proof of information loss.

supervision outside the world-model budget, so this is a test of recoverable information rather than unsupervised control sufficiency.

The policy interface matters. With an MLP torso, real pixels yield 0.91 walls per episode and frozen latents yield 0.94, with no discoveries in either group. An MLP on decoded latents similarly yields 0.92 walls. Thus failure of this MLP training recipe is not evidence that the encoder has lost the obstacle. These comparisons do not set a capacity limit for all MLP policies; they locate a limitation of the tested read-out and optimization procedure.

J.2 The tested imagination pipeline does not discover the strategy

In the earlier unmatched protocols, all three JEPA predictor variants produce poor downstream control (Table 12; per-seed outcomes in Appendix G). The original predictor yields 0.27 walls per episode; one residual sample and the mean of eight each yield 0.26. No policy among these 27 runs discovers the strategy; the highest final survival is 3.1%. Real-transition PPO on the same frozen encoders yields 0.94 walls. The difference measures the combined effect of learned transitions, fitted heads, and the changed PPO protocol. Even the stronger real-latent MLP result falls far short of reliable survival.

The flow reference supports feasibility at this task and nominal model budget: one of its nine policies discovers the strategy. The remaining eight pass 0.72–1.18 walls per episode. Since all nine use one dynamics fit with a different architecture and supervision, this reference does not establish a model-family ranking. Training the gate JEPA for twice and four times the steps also leaves imagined-policy survival at 0–3% in the tested extensions (encoder seed 0, policy seed 0, each predictor variant).

J.3 Prediction and fitted heads provide distinct failure signals

The same decoder that renders real encoded walls removes them from original predicted rollouts. Across all three encoder seeds, decoded shared-wall intensity is at most 0.02 by the second predicted step (Appendix G). The shared wall pixels occur in both possible gap futures, so a pixelwise mixture of those futures would retain them. This observation rules out that simple pixel-space picture; it does not determine what averaging or distribution shift does in the learned latent space.

Table 12: **Final gate outcomes, with predictor variants kept separate.** Mean ± sample standard deviation across policy runs; discovery is final survival above 50%. JEPA imagination variants and real-latent conditions each contain 3 models × 3 policies; pixel controls have three policies. Real training changes rewards, terminations, and PPO protocol alongside dynamics. Flow is a single-model reference.

Training / observation	Predictor or torso	Walls	Discovery
Imagined JEPA	Original	0.27 ± 0.22	$0/9$
	Residual sample	0.26 ± 0.32	$0/9$
	Residual mean of 8	0.26 ± 0.30	$0/9$
Real / frozen JEPA	MLP	0.94 ± 0.04	$0/9$
Real / pixels	MLP	0.91 ± 0.08	$0/3$
Real / decoded JEPA	CNN	1.57 ± 1.26	$2/9$
Real / pixels	CNN	2.91 ± 1.65	$2/3$
Imagined flow	Sample	1.45 ± 1.23	$1/6$
	Mean of 8	0.88 ± 0.14	$0/3$

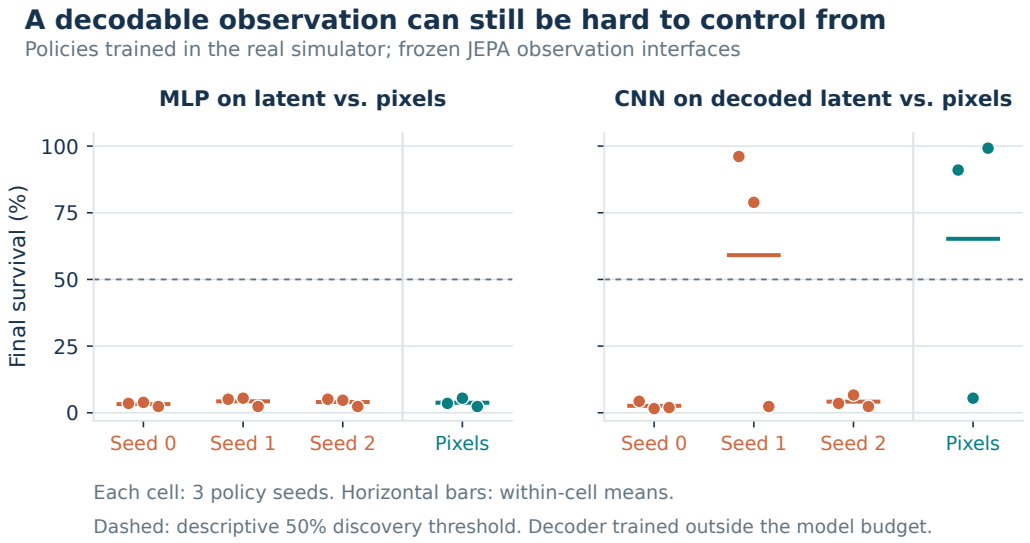


Figure 7: **Recoverable information can support successful control.** Final real-gate survival for MLP policies on frozen latents or pixels (left), and CNN policies on decoded latents or pixels (right). Points are policy runs, grouped by encoder seed; the dashed line is the 50% discovery criterion. Both decoded-latent discoveries use encoder 1. Reconstruction supervision and CNN training are additional to representation fitting.

A within-encoder counterfactual diagnostic makes the scale of the feature explicit. On 64 pre-reveal contexts for seed 0, the first-step squared distance between wall-present and wall-absent encodings is 0.035, while total teacher-forced prediction error is 0.145. The same ordering holds for all three seeds (Appendix G). This comparison concerns sensitivity to wall removal versus error in all features. It neither identifies a causal mechanism nor permits comparing information content through raw distances across independently fitted latent spaces.

The fitted heads expose another limitation. For encoder 0, termination AUC is 0.96 for border deaths but 0.65 for wall deaths. Predicted rewards for the two death causes are -0.89 and -0.05, respectively, against the true -1. An imagined policy therefore trains on a weak wall-collision signal. The residual-sampling variants retain these heads and the policy protocol. Their poor control outcomes therefore test this particular intervention, not distributional prediction in general.

K MATCHED-CONTROL PROTOCOL AND OUTCOMES

The primary matrix contains three fixed gate model checkpoints, five policy seeds (0–4), and three transition/signal conditions. Within each model/policy block, initialization, sampled starts, source identities, checkpoint hash, data hash, requested updates, policy parameter count, evaluation seed, and episode count must match. Start arrays are compared by semantic content hashes. Only complete three-condition blocks with 150 finished updates, finite required training diagnostics, and scalar outcomes reproduced from saved episode arrays enter the complete analysis. Missing, partial, or invalid runs retain their place in the planned denominator.

The primary endpoints are wall exits before or on the terminal transition and survival through the 128-step cap. A same-step wall exit counts even if that transition is terminal. A death on step 128 is counted as death, using episode return rather than duration. Survival above 50% is a secondary operational threshold, not independent evidence of a specific learned maneuver. Each saved final policy is evaluated on 256 real episodes with a common key schedule. Outcomes are paired within model and policy seed. Raw hidden state is used only to reconstruct the simulator transition source and must reproduce sampled current pixels exactly before training proceeds.

Within a fixed model, policy-seed effects are equally weighted. The main descriptive interval uses 10,000 paired-policy bootstrap resamples with analysis seed 7341, conditional on the evaluation set. A separate interval resamples evaluation slots jointly across policies while holding fitted policies fixed. It measures evaluation variation, not policy-training uncertainty. Across model fits, we show all three effects, their equal-model mean, and range; we do not treat 15 policy pairs as 15 independent models or report a model-family interval. All intervals are unadjusted and descriptive.

Exact-simulator states become absorbing after true termination. When fitted signals are used, their learned continuation can remain positive there; post-terminal alive mass and weighted reward are reported as part of the intervention's support change.

K.1 COMPLETED PRIMARY OUTCOMES

All 45 prescribed MLP runs passed full-budget, pairing, checkpoint-identity, and per-episode outcome checks. Their primary outcomes appear in Table 13. Table 14 retains every predeclared wall-count contrast with its within-model policy-seed interval. No confidence interval here estimates variation over a model family.

Table 13: **The exact training environment does not make this PPO recipe solve the gate.** Final real-environment outcomes, mean $\pm$ sample standard deviation across five policy seeds per row. Each of three fitted models receives all three matched conditions. None of the 45 policies exceeds 50% survival. "Fitted" means the unchanged learned reward and termination heads; "true" means simulator signals.

Model	Transitions / signals	Walls per episode	Survival (%)
0	Learned / fitted	0.285 ± 0.265	0.94 ± 1.28
0	Exact / fitted	0.230 ± 0.366	0.62 ± 0.98
0	Exact / true	0.452 ± 0.099	0.47 ± 0.33
1	Learned / fitted	0.217 ± 0.356	0.78 ± 1.35
1	Exact / fitted	0.068 ± 0.152	0.16 ± 0.35
1	Exact / true	0.341 ± 0.210	0.39 ± 0.39
2	Learned / fitted	0.320 ± 0.196	0.39 ± 0.28
2	Exact / fitted	0.096 ± 0.119	0.00 ± 0.00
2	Exact / true	0.543 ± 0.119	1.41 ± 0.45

L NATIVE-GRID POLICY SENSITIVITY

The sensitivity uses the same frozen models, rollout protocol, and three transition/signal conditions, with three policy seeds per model. The policy applies a CNN to the existing $4 \times 4 \times 32$ topology

Table 14: Matched primary wall-count differences. Intervals are 95% percentile bootstraps over five paired policy seeds within one fixed model, conditional on its evaluation set. They are unadjusted descriptive intervals; all three prespecified contrasts are retained.

Model	Contrast	Δ walls	95% interval
0	Exact/fitted − learned/fitted	-0.055	$[-0.368, +0.259]$
0	Exact/true − exact/fitted	$+0.222$	$[-0.109, +0.452]$
0	Exact/true − learned/fitted	$+0.166$	$[+0.015, +0.348]$
1	Exact/fitted − learned/fitted	-0.149	$[-0.492, +0.045]$
1	Exact/true − exact/fitted	$+0.273$	$[-0.027, +0.498]$
1	Exact/true − learned/fitted	$+0.123$	$[-0.195, +0.437]$
2	Exact/fitted − learned/fitted	-0.223	$[-0.429, +0.046]$
2	Exact/true − exact/fitted	$+0.447$	$[+0.329, +0.598]$
2	Exact/true − learned/fitted	$+0.223$	$[+0.034, +0.420]$

and has 583,236 parameters. All 27 runs complete 150 updates and 1,228,800 requested transitions. Nine three-arm start-array pairs, model identities, frozen source and policy hashes, and every scalar endpoint are verified from saved artifacts.

Table 15: Native-grid CNN sensitivity, kept separate from the primary MLP campaign. Each cell is mean $\pm$ sample standard deviation across three policies from one fixed model. None of the 27 policies exceeds 50% survival.

Model	Transitions / signals	Walls per episode	Survival (%)
0	Learned / fitted	0.000 ± 0.000	0.00 ± 0.00
0	Exact / fitted	0.431 ± 0.436	2.08 ± 1.85
0	Exact / true	0.616 ± 0.090	1.17 ± 0.39
1	Learned / fitted	0.284 ± 0.246	0.65 ± 0.60
1	Exact / fitted	0.738 ± 0.067	2.34 ± 0.68
1	Exact / true	0.367 ± 0.238	0.52 ± 0.45
2	Learned / fitted	0.202 ± 0.158	0.00 ± 0.00
2	Exact / fitted	0.229 ± 0.370	0.39 ± 0.68
2	Exact / true	0.452 ± 0.256	0.65 ± 0.81

Within-model exact/true minus learned/fitted wall-count effects are 0.616, 0.083, and 0.250; the corresponding paired-policy bootstrap intervals are $[0.516, 0.691]$, $[-0.312, 0.395]$, and $[-0.148, 0.477]$. These coarse intervals use only three paired policy seeds and condition on each checkpoint and its fixed evaluation set. They are not pooled with primary MLP intervals.

M FORECAST-READOUT PROTOCOL AND OUTCOMES

The three gate models stay frozen. Up to 16,384 training contexts and 8,192 evaluation contexts are sampled without replacement with seeds 20260919 and 20260920, using matching context IDs across models. Past context may not cross a reset. Forecasts use the original recorded outgoing actions, with horizons 1, 2, 4, and 8. Parameters and BatchNorm statistics must remain unchanged during extraction, and cached encodings are checked against the public rollout implementation.

The three feature domains are real future encoding, autoregressive forecast, and copied current encoding. Each domain has 512 coordinates. An affine ridge read-out uses mean squared error plus $10^{-3}\|W\|^2$, with unpenalized intercept and no hyperparameter search. Both declared scaler analyses are retained: current-training-latent mean/standard deviation shared across domains; and source-domain training mean/standard deviation for each horizon. Both use a fixed 10^{-6} scale floor, and every transfer applies its source-fitted scaler. No evaluation data determine a scaler, regularizer, checkpoint, or reporting condition.

Cube coordinates are divided by 63 for fitting and reported in pixels. Wall position is fitted and evaluated only on truth-present rows; the absent-wall sentinel is never regressed as a coordinate. Presence reports class prevalence, AUROC, average precision, and threshold accuracy/balanced accuracy, with undefined one-class metrics omitted. Every task includes a constant training-label-mean baseline and sample counts. A task with fewer than 64 fitting rows is marked insufficient.

The archive stores reset frames after terminal transitions, so physical-state read-outs use a horizon-specific continuing-episode mask. In particular, one-step evaluation does not select for eight-step survival. The hidden-gap negative control instead retains every pre-reveal context, including later resets, because future-survival selection could reveal the gap through the recorded actions. The gap label refers to the original current wall. Chance forecasting of this unavailable bit is compatible with correct treatment of uncertainty.

All fitted weights, scalers, selected IDs, labels, masks, per-example predictions, and provenance hashes are saved. Overlapping windows are not independent samples; any uncertainty uses evaluation stream or episode clusters. A weak read-out remains consistent with information loss, poor conditioning, limited support, or insufficient probe capacity.

Execution and uncertainty. All three models use the same canonical padded frame batch and predictor batch. A strict parity check against the public rollout function is bit-exact at the production shape, and model parameters and BatchNorm statistics remain unchanged. This canonicalization was fixed after an initial smoke parity failure and before any successful read-out fit; no tolerance, model weight, or metric was changed. The analysis recomputes 1,080 metric rows and 720 paired contrasts from saved float32 predictions without refitting. Intervals resample the 100 evaluation environment streams jointly across compared read-outs, with 2,000 replicates and seed 20260921. They condition on the fitted model and probe. AUROC and average precision are reported as point estimates.

For the model-2, horizon-eight pre-reveal cohort, the forecast-versus-copy MAE difference is -0.63 pixels with interval $[-1.41, 0.13]$ under source-specific scaling. The transfer-versus-adapted difference is 3.85 pixels with interval $[3.48, 4.22]$. Common scaling gives -0.64 $[-1.42, 0.12]$ and 3.89 $[3.52, 4.25]$, respectively. These are descriptive conditional comparisons; the strong adaptation result occurs in one of the three fitted models.

M.1 Read-out errors and a post-hoc tracking audit

The predeclared absolute-error table is retained below for every fitted model and both scaling analyses. Its global training-mean baseline is evaluated on the selected pre-reveal slice; it is not a slice-specific optimal constant. A separate audit of saved predictions examined correlations and RMSE without refitting or modifying any prediction. The model-2 forecast-fitted probe has Pearson correlation 0.246 on the full wall-present cohort and -0.175 within the pre-reveal slice. Its full-cohort RMSE is 18.65 pixels versus 13.49 for the global training-mean constant. Thus the MAE improvement should not be described as reliable tracking or successful recovery of wall position. A subsequent train-only constant-baseline audit uses the same horizon and cohort masks, with 1,509 training and 716 evaluation rows in the pre-reveal slice. The training-cohort mean position is 29.433 pixels and the median is 30. Their evaluation MAEs are 4.769 and 4.642 pixels, respectively, improving on every highlighted model-2 feature read-out in that slice. The full-cohort training median has MAE 10.560, compared with 9.112 for the forecast-fitted probe, while the latter's RMSE remains worse than the mean constant. This is an explicitly post-hoc interpretation check, with no feature-probe refitting, clipping, latent rescaling, or alteration of the declared metric outputs.

N Triggered termination sensitivity

This follow-up was triggered by the primary experiment's monitored post-terminal weights. It adds one condition: exact transitions, true termination, and the original learned reward evaluated on the current latent and outgoing action. It reuses every corresponding reference arm's saved start indices, checkpoint and data identity, policy initialization, update budget, and evaluation keys. The MLP stratum adds five policy seeds for each of three models; the native-grid stratum adds three, for 24 additional runs. The original 72 policies are neither retrained nor relabeled.

Table 16: **Read-out fitting changes measured forecast error.** Eight-step wall-position MAE in pixels on 716 pre-reveal continuing contexts. R, F, and C denote real future, forecast, and copied-current latents; arrows indicate fitting and evaluation domains. Both train-only scaling analyses are shown. The global training-mean baseline has MAE 14.24; the post-hoc training-cohort mean and median have MAE 4.77 and 4.64. The larger refitting gain occurs in model 2; all three models are retained.

Model	Scaling	R→R	R→F	F→F	C→C
0	Common	11.66	13.79	13.99	10.77
0	Source	11.37	13.77	13.99	10.77
1	Common	13.00	14.21	13.93	11.28
1	Source	12.82	14.44	13.93	11.28
2	Common	6.62	10.05	6.16	6.79
2	Source	6.53	10.02	6.16	6.79

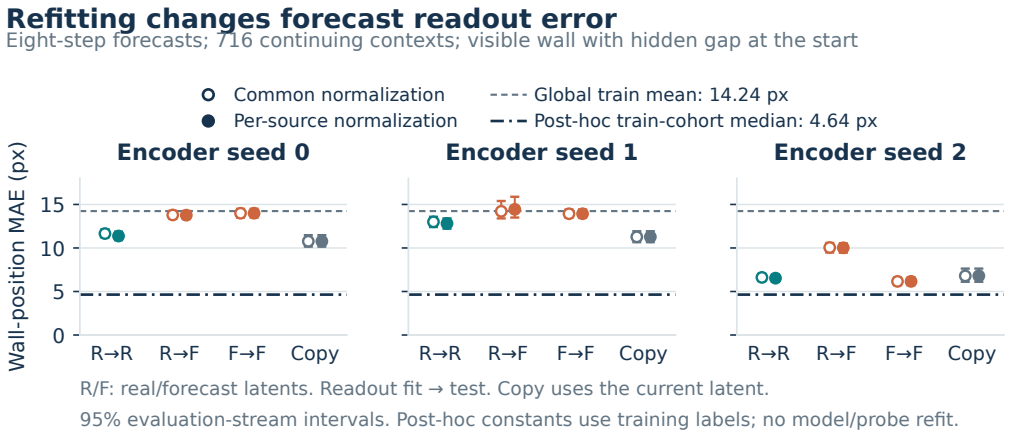


Figure 8: **Refitting changes forecast read-out error.** All three model seeds and both predeclared normalization schemes are retained. The original baseline is the global training-label mean evaluated on the selected slice; the added cohort-median constant is a post-hoc train-only check. Lower slice MAE can reflect calibration as well as tracking; the model-2 correlation audit and full-cohort RMSE prevent a reliable-position-recovery claim.

All added runs complete 150 updates. All 3,600 recorded training updates have finite required diagnostics and exactly zero continuation-weighted post-terminal mass and reward. Every final policy binary hash and per-episode endpoint is checked. Table 17 gives the new condition's outcomes. None of the added policies exceeds 50% survival.

Table 17: Exact transitions, true termination, and fitted reward: the added termination-sensitivity condition. Mean ± sample standard deviation over paired policy seeds within one fixed model and torso. The two torso strata remain separate.

Torso	Model	Policies	Walls per episode	Survival (%)
MLP	0	5	0.164 ± 0.239	0.62 ± 0.90
MLP	1	5	0.216 ± 0.244	0.86 ± 1.05
MLP	2	5	0.078 ± 0.107	0.16 ± 0.21
Grid CNN	0	3	0.102 ± 0.110	0.65 ± 1.13
Grid CNN	1	3	0.000 ± 0.000	0.00 ± 0.00
Grid CNN	2	3	0.181 ± 0.313	0.65 ± 1.13

Table 18: Wall-count changes in the triggered sensitivity. "True termination" compares the new condition with exact transitions and fitted reward/termination. "True reward" compares exact transitions and true reward/termination with the new condition. Each entry gives the paired mean and its conditional 95% policy-seed bootstrap interval.

Torso	Model	True termination	True reward
MLP	0	-0.066 $[-0.486, +0.261]$	$+0.287$ $[+0.003, +0.512]$
MLP	1	$+0.148$ $[-0.083, +0.380]$	$+0.124$ $[-0.075, +0.391]$
MLP	2	-0.018 $[-0.162, +0.094]$	$+0.465$ $[+0.312, +0.619]$
Grid CNN	0	-0.329 $[-0.785, +0.000]$	$+0.514$ $[+0.422, +0.605]$
Grid CNN	1	-0.738 $[-0.793, -0.664]$	$+0.367$ $[+0.117, +0.590]$
Grid CNN	2	-0.048 $[-0.656, +0.543]$	$+0.271$ $[+0.164, +0.453]$

The termination contrast changes continuation weighting and removes the learned after-death extension. The reward contrast holds exact transitions and true termination fixed; policies trained under different rewards can then visit different states. The latter increases the mean wall count in all six cells, while survival changes have mixed signs. Both comparisons condition on the saved encoder/head checkpoint and the given policy/training procedure. The 10,000-resample intervals are unadjusted and descriptive; three model fits do not establish a model-family effect.

O VISUAL-GOAL CONTROL IN THE DETERMINISTIC GRAY-NOISE WORLD

All three original `gray_frame_still` grid models use the same 128 freshly generated cases and unchanged planning settings. Evaluation seed 20260920 produces native resets followed by seven random actions; a history is admitted only if it survives this past prefix. No candidate is rejected according to a future model result. The goal is sampled uniformly from safe positions on the same three-pixel action lattice, with shortest Manhattan action distance between six and 12. Mean initial L1 distance is 26.883 pixels, with range 18–36. The goal image has independently drawn gray noise and remains fixed within its case.

The planner sees genuine eight-frame histories and the RGB goal. Horizon-three MPC enumerates all 64 native action sequences and minimizes summed squared latent-goal distance over the three predictions. It executes the first action, observes the true successor, and replans, for at most 24 real steps. Horizon-one MPC is a predeclared comparison. No decoder, position probe, reward head, altered reward, latent rescaling, or privileged state enters either learned planner. History and goal encoding use a common fixed $[16, 8, 63, 63, 3]$ batch shape. Outgoing action alignment is $[a_0, \ldots, a_6, a_{\text{candidate}}]$.

The primary metric is a first exact goal hit before native death. Execution stops on the first hit or death, and all 128 cases remain in the denominator. Secondary metrics are minimum/final L1 distance, entry within three pixels, deaths, and first-hit time. Random and blind-cycle references share execution noise with each model. The privileged exact-transition reference searches the same horizon with physical squared-distance costs and rejects plans with intermediate death. It reaches every goal in the shortest number of actions; this is a protocol check, not a learned competitor.

Horizon-three exact-hit counts are 5, 5, and 9 for models 0, 1, and 2. Their minimum-distance means are 22.242, 22.945, and 22.031 pixels, against 19.406 for random actions. Native-death counts are 9, 8, and 24, against 20 for random. Horizon-one hits no goal for any model, with minimum-distance means 23.906, 24.797, and 24.047 pixels and death counts 4, 6, and 5. The fixed cycle survives but reaches no goal. These outcomes establish failure of this short-horizon latent-cost recipe on the specified distant-goal distribution; they are not a positive planning control.

Independent reconstruction checks all 1,024 prefix frames per model, every action and native transition, reachability, exact goal draws, first-hit/death absorption, summaries, visual episode selection, and recorded hashes. Across the three fits, all cohort arrays and reference trajectories are bit-identical. The three model evaluations do not create 384 independent cases. Visual episodes 0, 1, and 2 were fixed before results, and decoded selected forecasts are marked invalid after terminal

padding. The empty arena permits reactive goal movement; the task does not demonstrate a need for deep planning.

Table 19: **The original distant-goal test remains a weak result.** Initial goals are six to 12 native actions away. The same 128 goals and histories are used for all three models and references. Distances are mean L1 pixels over all cases; lower is better. Random, blind-cycle, and privileged reference trajectories are shared across model evaluations.

Controller	Exact hits	Minimum distance	Final distance
JEPA H3, model 0	5/128	22.24	27.09
JEPA H3, model 1	5/128	22.95	27.75
JEPA H3, model 2	9/128	22.03	28.55
Random	3/128	19.41	30.73
Blind cycle	0/128	24.59	26.88
Privileged reference	128/128	0.00	0.00

O.1 Separately declared distance-response follow-up

The distance-response experiment was declared after the failed 6–12-action primary result. It characterizes the same controller's operating range and does not replace that primary. All six exact goal-distance shells—one, two, three, six, nine, and 12 native actions—use the same 128 original valid histories and all three existing model seeds. The H3 and H1 planners, 24-action budget, random/blind-cycle/privileged references, native dynamics, and encoder/predictor execution shapes remain fixed. No planner or representation parameter changes.

A newly fixed goal seed, 20260921, samples uniformly among safe reachable positions on each exact Manhattan shell. The RGB goal background is shared across the six goals for a prefix, so only object position changes. Execution noise and random actions are paired by original candidate ID. The outcome is exact first hit before death. Every model–distance cell retains its own 128-case count and Wilson interval; the 768 correlated rows are never pooled into an inferential interval. The within-three-pixel secondary is already satisfied initially on the one-action shell and is not evidence of movement there.

Case 0 at every shell is the predeclared visualization. Its full recorded native trajectory is re-rendered with the recorded actions and execution keys; padding freezes after first hit or death. The initial H12 CEM alternative was assessed but not launched. Thus any distance trend concerns the unchanged H3 controller over different requested goals, not a tuned-horizon comparison.

All three models and the final independent audit completed. The H3 hit counts at distances $(1, 2, 3, 6, 9, 12)$ are $(128, 128, 121, 25, 3, 0)$, $(128, 128, 123, 32, 2, 0)$, and $(128, 128, 124, 34, 1, 0)$ for seeds 0, 1, and 2, respectively. Table 20 keeps the H1 comparison and paired references alongside these outcomes. The independent audit checks 8,179 conditions and reconstructs 176,334 executed native transitions from raw artifacts.

P Exact variance omission in two frozen encoders

This calculation uses the existing seed-0 grid JEPA in black-background wind as its declared primary condition and persistent-gray wind as its declared secondary condition. Their independently measured real-encoding position L1 errors are 0.126 and 0.717 pixels, respectively. Those diagnostics motivate an encoder with accessible state; no probe, learned transition model, task head, or policy is used in the variance calculation.

Fixed transition law and goals. A fresh collection uses 128 streams and 512 transitions per stream, with environment seed 20260923. From 46,136 past-valid eight-frame contexts, 1,024 are sampled without replacement using seed 20260924. Eligibility for the final calculation requires every action/innovation branch to remain inside the safe region and before the time cap. This admits the same 843 contexts in both worlds; 181 are excluded for border risk and none for the cap. Every



Figure 9: **Actual paths across the declared goal-distance shells.** Model seed 0, fixed case 0 in every shell. Curves are executed simulator positions, annotated only for evaluation; the planner receives RGB history and an RGB goal. Paths stop at the first exact hit, native death, or 24 actions. The examples and all six shells were fixed before the distance-response outcomes.

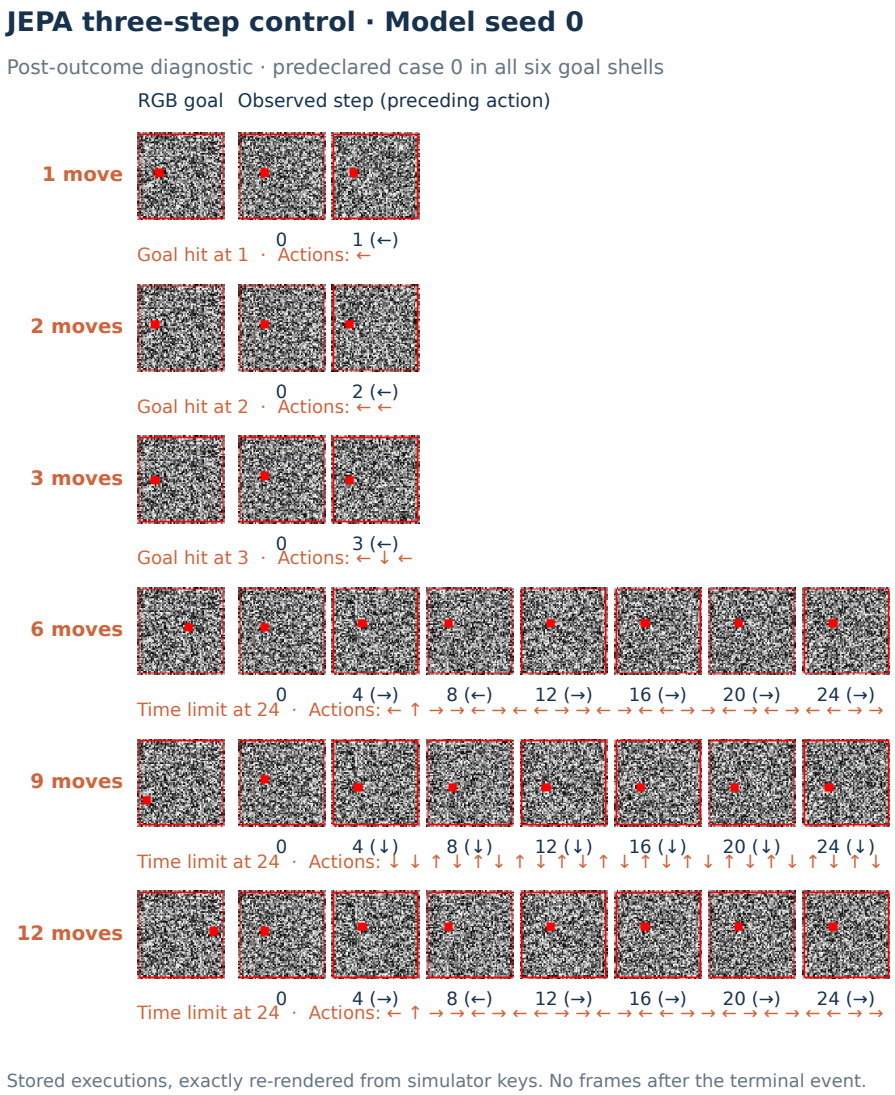


Figure 10: Native-color RGB views of the same preselected goal-distance executions for model 0. These are exact keyed re-renderings of recorded real trajectories, with no model forecast or invented continuation. The full artifact includes the corresponding displays for models 1 and 2.

Table 20: Distance-response exact-hit counts. Initial distance is measured in native three-pixel cardinal moves. Every cell has denominator 128. The same six shells, prefixes and controller settings are retained for all models; no pooled binomial interval is used.

Distance	JEPA H3, model			JEPA H1, model		
	0	1	2	0	1	2
1	128	128	128	128	128	128
2	128	128	128	39	36	46
3	121	123	124	22	16	21
6	25	32	34	4	4	1
9	3	2	1	0	0	0
12	0	0	0	0	0	0

Distance	Random	Blind cycle	Privileged reference
1	65	67	128
2	46	0	128
3	19	0	128
6	5	0	128
9	1	0	128
12	0	0	128

admitted context is crossed with all 25 goals in $\{8, 18, 28, 38, 48\}^2$, giving 21,075 pairs per checkpoint. Goal images use the context's episode texture. No goal or context is selected according to model costs.

Each coordinate of the wind increment takes $-1, 0, +1$ with probabilities $0.1, 0.8, 0.1$. All nine joint labels retain their original product probabilities, including duplicate clipped wind outcomes. The simulator state and texture are used to enumerate the exact next images. Thus the resulting mean is conditioned on the full physical state and texture, not assumed equal to the mean available from a learned latent history. Current wind is independently verified from the preceding two positions and action.

Cost decomposition and action comparison. For a fixed encoder f, goal z_g, and action a, compute the nine encoded successors $z'_{a\delta}$, their exact weighted mean μ_a, and

$$M_a = \frac{\|\mu_a - z_g\|^2}{D}, \qquad J_a = \frac{\sum_\delta q_\delta \|z'_{a\delta} - z_g\|^2}{D} = M_a + \frac{\mathrm{trCov}(Z' \mid s, a)}{D}, \qquad D = 512. \tag{11}$$

All arithmetic after encoding uses float64. The independent direct-versus-decomposed calculation has maximum residual 1.33×10^{-15} and 1.67×10^{-16} in the two checkpoints. Tolerance-optimal action sets use the fixed threshold $10^{-10} \max(1, \max_a |M_a|, \max_a |J_a|)$. The primary disagreement requires disjoint optimal sets. Regret is the minimum exact-cost excess among mean-cost minimizers, so arbitrary tie breaking cannot produce the finding. All pairs, including zero-regret pairs, enter the averages.

Table 21: Exact mean-only versus expected squared latent-goal scoring. Intervals resample complete collection streams, conditional on the frozen checkpoint. Latent regret uses coordinate-mean squared units and is not comparable as a physical error across independently fitted encoders.

Condition	Action disagreement (%)	Mean latent regret
Black background + wind	2.65 [2.19, 3.15]	7.50×10^{-5}
Persistent gray + wind	1.08 [0.83, 1.36]	3.80×10^{-6}

Regret intervals are $[5.77, 9.63] \times 10^{-5}$ and $[2.02, 6.86] \times 10^{-6}$, respectively. Mean-minus-exact-latent selected actions change expected physical squared-goal cost by -0.169 pixels squared with interval $[-1.107, 0.666]$ and $+0.200$ with interval $[-0.433, 0.961]$. Both include zero; no physical-control benefit is established. The discrepancy is a modest but direct consequence of omitting vari-

ance under the specified latent metric. For this one-step quadratic objective, a mean plus the scalar covariance trace is sufficient; a full generative model is not logically necessary.

Controls and uncertainty. Identity-position, fixed affine-position, and deterministic-branch controls all have zero robust action disagreements. For physical position, conditional wind covariance is action independent, so its variance term cannot change the one-step squared-goal ranking. The learned nonlinear encoder can make latent variance action dependent. Canonical padded inference uses 2,048 frames, with bit-exact context and permutation parity and unchanged parameters/BatchNorm statistics. The simulator checks all sampled transitions lie in enumerated support and that all safe one-step rewards are zero.

Goals are averaged within context, then 2,000 bootstrap samples resample full collection streams; 127 streams contain admitted contexts. The intervals quantify context sampling for each fixed encoder, not training-seed uncertainty. All-branch safety selects an interior cohort and excludes terminal decisions. This experiment isolates mean-only latent-goal scoring, not survival, representation sufficiency for arbitrary tasks, learned-predictor accuracy, or recursive mean error.